

# Performance and Robustness Benchmarking of Battery SOC Estimation Methods for Embedded Applications

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## Abstract

Robustness under disturbed sensing and signal-integrity faults is central to battery State-of-Charge (SOC) estimation. This study benchmarks four representative SOC-estimation approaches under one causal and cell-disjoint protocol: a direct-measurement model (DM) implemented as fixed-capacity Coulomb counting, a hybrid direct-measurement model (HDM) combining Coulomb counting with data-driven SOH adaptation, a hybrid equivalent-circuit model (HECM) based on a second-order RC observer, and a data-driven neural model (DD) implemented as a recurrent estimator. LFP aging trajectories cover multiple load, thermal, and SOH conditions. The evaluation comprises two complementary branches. The hardware performance benchmark verifies numerical equivalence between the software and microcontroller implementations and measures inference latency, flash occupancy, and peak runtime RAM. The robustness benchmark characterizes estimator behavior in three dimensions. Accuracy quantifies agreement with the reference SOC under nominal input conditions. Robustness quantifies the stability of the SOC estimate and its resistance to error growth when measurements are corrupted or unavailable. Recovery quantifies how rapidly a reliable SOC estimate is re-established after a defined initialization mismatch. Nominal cell-macro MAE is 0.0690 for DM, 0.0412 for HDM,

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0.0337 for HECM, and 0.0258 for DD. The DD representative achieves the lowest nominal error and leads several disturbance cases, whereas HECM provides the strongest observed recovery profile. Individual scenarios therefore produce different model rankings. On the STM32H753, all four isolated SOC implementations reproduce their software references, remain within the available memory, and execute within the one-second sampling interval. Their latency and memory requirements differ substantially, demonstrating that deployment cost must be considered alongside estimation quality. The results show that nominal accuracy alone provides an incomplete basis for model selection and must be complemented by robustness, recovery, and embedded feasibility.

*Keywords:* Battery management system, State of charge, Robustness benchmark, Fault tolerance, Equivalent circuit model, Recurrent neural network, Embedded implementation, Microcontroller

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## 1 Nomenclature

### 2 *Acronyms*

|    |             |                                 |
|----|-------------|---------------------------------|
| 3  | <b>BMS</b>  | Battery management system       |
| 4  | <b>SOC</b>  | State of charge                 |
| 5  | <b>SOH</b>  | State of health                 |
| 6  | <b>DM</b>   | Direct-measurement model        |
| 7  | <b>HDM</b>  | Hybrid direct-measurement model |
| 8  | <b>HECM</b> | Hybrid equivalent-circuit model |
| 9  | <b>DD</b>   | Data-driven neural model        |
| 10 | <b>ECM</b>  | Equivalent circuit model        |
| 11 | <b>EKF</b>  | Extended Kalman filter          |
| 12 | <b>MAE</b>  | Mean absolute error             |
| 13 | <b>RMSE</b> | Root-mean-square error          |
| 14 | <b>ADC</b>  | Analog-to-digital converter     |
| 15 | <b>CV</b>   | Constant-voltage phase          |

|    |                              |                                                         |
|----|------------------------------|---------------------------------------------------------|
| 16 | <b>OCV</b>                   | Open-circuit voltage                                    |
| 17 | <b>GRU</b>                   | Gated recurrent unit                                    |
| 18 | <b>LSTM</b>                  | Long short-term memory                                  |
| 19 | <b>LFP</b>                   | Lithium iron phosphate                                  |
| 20 | <i>Symbols</i>               |                                                         |
| 21 | $t$                          | Time                                                    |
| 22 | $I(t)$                       | Current signal at time $t$                              |
| 23 | $U(t)$                       | Voltage signal at time $t$                              |
| 24 | $T(t)$                       | Temperature signal at time $t$                          |
| 25 | $\widehat{\text{SOC}}(t)$    | Estimated state of charge at time $t$                   |
| 26 | $\widehat{\text{SOH}}(t)$    | Estimated state of health at time $t$                   |
| 27 | $\text{SOC}_{\text{ref}}(t)$ | Dataset SOC ground truth at time $t$                    |
| 28 | $\text{SOH}_{\text{ref}}(t)$ | Reference SOH target at time $t$                        |
| 29 | $b_I$                        | Multiplicative current-gain error                       |
| 30 | $\Delta U$                   | Voltage bias or offset                                  |
| 31 | $\Delta T$                   | Temperature bias or offset                              |
| 32 | $\sigma_I$                   | Standard deviation of injected current noise            |
| 33 | $\sigma_U$                   | Standard deviation of injected voltage noise            |
| 34 | $\sigma_T$                   | Standard deviation of injected temperature noise        |
| 35 | $Q_c$                        | Online cumulative charge state used by the SOC models   |
| 36 | $C_{\text{nom}}$             | Nominal capacity                                        |
| 37 | $C_{\text{eff}}$             | Effective capacity used inside the estimator            |
| 38 | $U_{\text{max}}$             | Upper voltage threshold used for full-charge detection  |
| 39 | $U_{\text{tol}}$             | Voltage tolerance used in the full-charge condition     |
| 40 | $U_1, U_2$                   | Polarization voltages of the first and second RC branch |
| 41 | $\Delta\text{MAE}$           | Error increase relative to baseline                     |

## 42 1. Introduction

### 43 1.1. Motivation and practical relevance

44     Reliable battery state estimation is a core BMS function because SOC and  
45 SOH determine safety margins, usable energy, operating limits, charge con-  
46 trol, and degradation-aware system management in both mobile and stationary  
47 lithium-ion battery applications [1, 2, 3]. Practical deployment quality is not  
48 defined by clean-condition accuracy alone, because real BMS operation is ex-  
49 posed to sensor gain and offset errors, additive noise, initialization uncertainty  
50 after restart, missing samples, irregular sampling, communication interruptions,  
51 and transient signal spikes caused by measurement-chain limitations or distur-  
52 bances in the surrounding system [4, 5, 6]. The relevant engineering question  
53 is therefore not only whether an estimator achieves low MAE under nominal  
54 measurements, but also whether it converges back to a reliable state after dis-  
55 turbed initialization and whether it remains stable under physically plausible  
56 measurement corruption.

57     As summarized conceptually in Figure 1, the present benchmark treats  
58 deployment-facing estimator quality as a three-dimensional problem consisting  
59 of nominal accuracy, recovery after state inconsistency, and robustness against  
60 disturbed measurements and signal-integrity faults, embedded within broader  
61 constraints on memory and execution time. The central question is how four  
62 structurally different SOC-estimation classes, represented by the implementa-  
63 tions studied here, respond to common failure mechanisms and microcontroller  
64 constraints, and whether their nominal ranking remains informative under those  
65 conditions. This distinction is essential because one implementation can be ac-  
66 curate under clean measurements yet difficult to re-anchor, sensitive to a specific  
67 input fault, or too costly under its trained inference semantics.

### 68 1.2. State of the art in battery state estimation

69     The literature commonly groups battery-state estimators into direct-measurement  
70 or integration-based approaches, model-based approaches based on equivalent-  
71 circuit or electrochemical models, hybrid approaches that combine physical

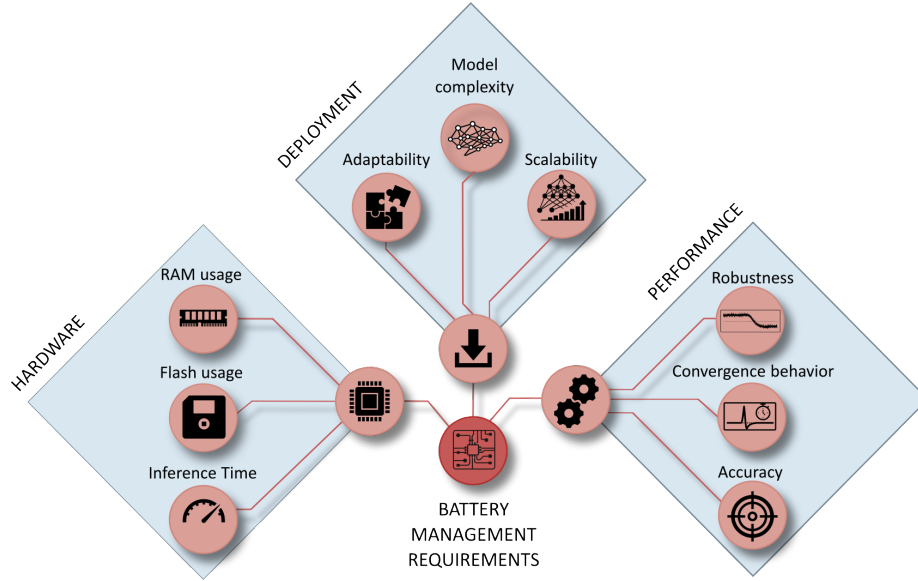


Figure 1: Conceptual overview of deployment-oriented requirements for embedded BMS state estimation. The broader requirement space includes implementation effort, hardware constraints, and estimator performance, whereas the present study focuses on the performance branch and evaluates baseline accuracy, convergence behavior, and robustness under measurement disturbances.

72 structure with learned components, and purely data-driven models based on  
 73 neural sequence learning [7, 1, 2, 3]. Direct-measurement approaches, typically  
 74 based on Coulomb counting, are attractive because they are simple, compu-  
 75 tationally inexpensive, and easy to deploy. However, they remain structurally  
 76 sensitive to current-gain error, missing data, and initialization errors because  
 77 they lack an internal correction mechanism [7, 4]. Model-based approaches,  
 78 especially ECM- and observer-based estimators, offer physical interpretability  
 79 and voltage-based correction capability, but their practical behavior depends  
 80 strongly on the quality of model parametrization, operating-point observability,  
 81 and measurement quality [8, 2, 4].

82 Hybrid estimators seek to combine physical state propagation with learned  
 83 adaptation of capacity, parameters, or auxiliary states, which has made joint  
 84 SOC/SOH estimation an active research direction [9, 10, 11]. This combination

can improve baseline calibration, while it does not necessarily eliminate structural sensitivity to measurement disturbances. Data-driven approaches based on recurrent or sequence models can exploit nonlinear cross-relations between current, voltage, temperature, and derived online features, and recent work has also demonstrated their embedded feasibility through TinyML-oriented implementations, pruning, and quantization [12, 13, 14, 15, 3].

### 1.3. Performance and embedded deployment gap

A large share of the SOC estimation literature still emphasizes clean-data accuracy, average error metrics, or computational efficiency, while comparatively fewer studies analyze how estimators behave under realistic measurement faults and timing corruption [1, 13, 14, 15]. Existing robustness studies often remain confined to a single model family, for example observer-based robustness against modeling and measurement errors [4] or robust estimation concepts for specific BMS architectures under noise and uncertainty [5], whereas cross-family comparisons under the same disturbance protocol and under comparable embedded constraints remain limited. Benchmarking studies in the broader BMS context underline the importance of reproducible scenario design and deployment-relevant validation, but they do not by themselves close the gap between clean-data estimator comparison and measurement-level performance benchmarking across fundamentally different estimator classes [6]. This gap is especially relevant in embedded BMS design, where estimator selection must jointly consider nominal accuracy, convergence, robustness under disturbed measurements, and computational constraints.

### 1.4. Contribution and study objectives

This work addresses the gap through three linked questions: whether clean-data accuracy predicts performance under measurement and signal-integrity disturbances, which structural mechanisms govern degradation and recovery, and how the resulting behavior trades against embedded execution cost. To answer them, four deployment-oriented representatives are executed under one

causal protocol on cell-disjoint LFP aging data covering multiple load and SOH states. The campaign combines physically interpretable measurement, timing, availability, quantization, and initialization disturbances with a common recovery criterion, cell-level uncertainty analysis, a paired SOH-input ablation, and STM32 measurements of the isolated SOC cores.

The four evaluated implementations provide distinct update and correction mechanisms whose behavior can be traced across the same disturbances, recovery criterion, and hardware measurements. The resulting comparison links estimator selection to the fault profile and resource constraints of the target BMS application.

## 2. Methodology

### 2.1. Estimator representatives and class boundaries

Figure 2 summarizes the causal benchmark chain, from measured signals and online feature reconstruction to disturbance injection and global and local evaluation. The four estimators were selected because their dominant SOC update and correction mechanisms differ. Table 1 relates these mechanisms to the broader design space and specifies the implementation evaluated for each class. Learned corrections, higher-order dynamics, richer sensing, and more complex sequence architectures extend these class envelopes while also changing deployment cost.

Table 1: Estimator-class envelope and implementation scope. Literature in the second column supports the defining mechanisms and broader capability ranges, while the last column specifies the representative evaluated in this study.

| <b>Class</b>              | <b>Defining mechanism and broader capability range</b>                                                                                                                                                                        | <b>Representative used in this study</b>                                                                             |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Direct measurement        | Causal integration of measured current using an assumed capacity. Extensions can include coulombic efficiency, capacity adaptation, OCV or end-point anchoring, plausibility logic, and reset thresholds [7, 13].             | Fixed-capacity Coulomb counting with a sustained full-charge CV reset and no voltage-residual correction             |
| Hybrid direct measurement | Integration core augmented by an estimated or learned state or parameter. Extensions can include learned capacity or efficiency, residual correction, multi-sensor fusion, and adaptive anchoring [10, 11].                   | Same Coulomb-counting and reset logic as DM, with effective capacity supplied by the shared causal LSTM-SOH estimate |
| ECM observer              | Coulomb-counted SOC propagation combined with a voltage model and feedback correction. Extensions can include higher RC order, hysteresis, thermal states, operating-point maps, and different nonlinear observers [7, 4, 2]. | Second-order RC model with EKF correction, charge/discharge SOC-SOH lookup surfaces, and shared causal SOH input     |

Table 1: Estimator-class envelope and implementation scope (continued).

| Class       | Defining mechanism and broader capability range                                                                                                                                                                          | Representative used in this study                                                                                    |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Data driven | Learned mapping from measured signals or their history to SOC. Implementations can use feed-forward, convolutional, recurrent, attention-based, physics-informed, raw-signal, or engineered-feature models [12, 15, 13]. | Structurally pruned rolling-window GRU using voltage, current, temperature, SOH, $Q_c$ , derivatives, and $\Delta t$ |

134 All SOH-dependent representatives, namely HDM, HECM, and DD, receive  
 135 the same causal LSTM-SOH trace so that differences arise from how the SOC  
 136 branch consumes that context. DM is the low-complexity integration reference.  
 137 HDM isolates the effect of learned capacity adaptation without adding volt-  
 138 age feedback. HECM adds model-based voltage-residual correction to current-  
 139 driven state propagation. DD represents compact temporal learning. The GRU  
 140 was selected because SOC and disturbance recovery depend on recent signal his-  
 141 tory while its gated single-state structure is lighter than a comparable LSTM.  
 142 Because DD also consumes the engineered  $Q_c$  feature, it is a data-driven se-  
 143 quence estimator with physics-informed inputs rather than a raw-signal-only  
 144 neural model.

## 145 2.2. SOC estimation methods

### 146 2.2.1. Direct-measurement model (DM)

147 The direct-measurement SOC formulation used in DM, HDM, and as an  
 148 explicit auxiliary feature in DD follows the same online cumulative-charge state,  
 149 denoted here uniformly as  $Q_c$ . It is updated causally from the measured current  
 150 stream as

$$Q_c(k) = Q_c(k-1) + \frac{I_k \Delta t_k}{3600 \text{ s h}^{-1}}, \quad (1)$$

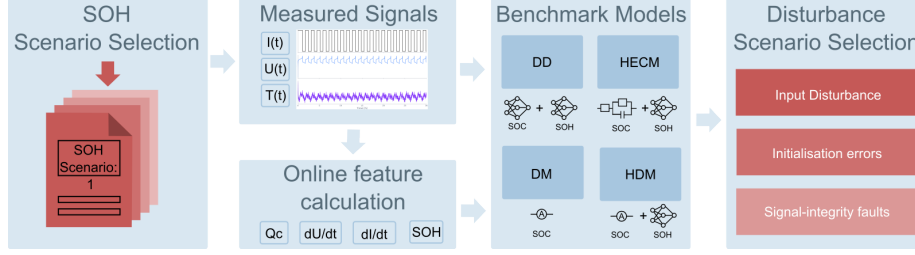


Figure 2: Methodology overview for the causal benchmark chain. Measured battery signals are converted into online auxiliary features, processed by representative estimator implementations, exposed to measurement and initialization interventions, and finally evaluated through global and local performance metrics.

151 where  $I_k$  is given in ampere and  $\Delta t_k$  in seconds, so that the factor  $3600 \text{ s h}^{-1}$   
 152 converts the increment to ampere-hours. The sign convention is fixed by the  
 153 benchmark implementation and the state is reset to  $Q_c = 0$  after a sustained  
 154 constant-voltage full-charge event. In the DM, this same charge state is mapped  
 155 directly to SOC through the fixed nominal capacity

$$\widehat{\text{SOC}}_{\text{DM}}(k) = \text{clip}\left(1 + \frac{Q_c(k)}{C_{\text{nom}}}, 0, 1\right). \quad (2)$$

### 156 2.2.2. Hybrid direct-measurement model (HDM)

157 The HDM preserves exactly the same Coulomb-counting core and differs  
 158 only in the capacity term. The SOH estimate rescales the usable capacity as

$$C_{\text{eff}}(k) = C_{\text{nom}} \cdot \widehat{\text{SOH}}(k), \quad (3)$$

159 which yields

$$\widehat{\text{SOC}}_{\text{HDM}}(k) = \text{clip}\left(1 + \frac{Q_c(k)}{C_{\text{eff}}(k)}, 0, 1\right). \quad (4)$$

160 This formulation makes the relation between DM and HDM explicit: both use  
 161 the same online charge state  $Q_c$ , while only the denominator changes through  
 162 online SOH support. In both direct-measurement variants, a sustained near-  
 163 maximum voltage condition is interpreted as a constant-voltage full-charge event.  
 164 If the voltage remains above  $U_{\text{max}} - U_{\text{tol}}$ , where  $U_{\text{tol}}$  denotes the admissible volt-  
 165 age margin below the upper threshold  $U_{\text{max}}$ , for the configured CV duration,

the internal charge state is reset so that  $Q_c = 0$  and therefore  $\widehat{\text{SOC}} = 1$ . This reflects the practical assumption that a completed CV phase provides a physically anchored full-charge reference.

### 2.2.3. Hybrid equivalent-circuit model (HECM)

The hybrid ECM model is implemented as a second-order RC model with the state vector

$$\mathbf{x}_k = \begin{bmatrix} \widehat{\text{SOC}}_k & U_{1,k} & U_{2,k} \end{bmatrix}^\top, \quad (5)$$

where  $U_1$  and  $U_2$  denote the polarization voltages of the two RC branches. The prediction step propagates the state according to

$$\mathbf{x}_k^- = \mathbf{A}_k \mathbf{x}_{k-1} + \mathbf{b}_k I_{k-1}, \quad (6)$$

with

$$\mathbf{A}_k = \text{diag}\left(1, e^{-\Delta t_k/\tau_1}, e^{-\Delta t_k/\tau_2}\right), \quad \mathbf{b}_k = \begin{bmatrix} \eta \Delta t_k / C_b \\ R_1(1 - e^{-\Delta t_k/\tau_1}) \\ R_2(1 - e^{-\Delta t_k/\tau_2}) \end{bmatrix}, \quad (7)$$

where  $C_b = C_0 \widehat{\text{SOH}}$  is the SOH-scaled available capacity. The corrected terminal-voltage estimate is written as

$$\hat{U}_k = \text{OCV}(\widehat{\text{SOC}}_k, \widehat{\text{SOH}}_k, I_k) + U_{1,k} + U_{2,k} + R_i I_k, \quad (8)$$

and the Kalman update uses the residual between  $\hat{U}_k$  and the measured terminal voltage to correct the internal state. In the present implementation, the state propagation is driven by the previous current sample  $I_{k-1}$ , whereas the measurement equation uses the current sample  $I_k$  available at the correction step. The ECM parameters are not constant. The implementation interpolates  $R_i$ ,  $R_1$ ,  $R_2$ ,  $\tau_1$ ,  $\tau_2$ , OCV, and dOCV/dSOC from a preidentified lookup table over SOC and SOH, with separate charge and discharge parameter surfaces. The HECM is implemented as a two-RC observer with SOH-dependent parameter scheduling.

#### 2.2.4. Data-driven SOC model (DD)

The data-driven SOC model is a GRU-based recurrent estimator with the feature vector  $\{U \text{ [V]}, I \text{ [A]}, T \text{ [}^\circ\text{C]}, \text{SOH}, Q_c, dU/dt, dI/dt, \Delta t\}$ . The benchmarked, structurally pruned and fine-tuned model consists of a single-layer GRU with hidden size 67 followed by an MLP head with one hidden layer of size 96 and a sigmoid output layer. During optimization, the recurrent core is exposed to causal sequence segments of length  $L_{\text{SOC}} = 2024$  samples. Within this feature set,  $Q_c$  is exactly the same online cumulative-charge state defined in (1). It is included explicitly so that the neural estimator receives the same charge-throughput information that also drives DM and HDM. The derivative channels are computed online as the finite differences  $dI/dt(k) = (I_k - I_{k-1})/\Delta t_k$  and  $dU/dt(k) = (U_k - U_{k-1})/\Delta t_k$ . They provide local transient information that is not captured by the absolute channels alone. The additional feature  $\Delta t_k$  allows the network to distinguish amplitude changes from sampling-interval changes under irregular timing. During the primary benchmark, each output is calculated from the latest causal 2024-sample window and recurrent state is reset between windows, matching training and validation. This configuration uses measured and causally reconstructed inputs up to the current sample. The hardware analysis additionally evaluates continuous hidden-state forwarding as a deployment variant and reports its accuracy and resource demand separately.

#### 2.3. SOH estimation method

The shared SOH estimator is an LSTM-based sequence-to-sequence model that operates on hourly aggregated online features derived from  $\{U \text{ [V]}, I \text{ [A]}, T \text{ [}^\circ\text{C]}, \text{EFC}, Q_c\}$ , where the aggregation uses the statistics mean, standard deviation, minimum, and maximum for each feature [16, 17, 18, 19]. This transforms the five base channels into a 20-dimensional hourly feature vector. Architecturally, the SOH network first projects the aggregated input through a two-layer feature embedding block with size 128, layer normalization, GELU activations, and dropout, then processes the sequence with a two-layer unidirectional LSTM of hidden size 112, and finally maps the recurrent output through

three residual MLP blocks and a prediction head with hidden size 160 to obtain the SOH estimate for each hourly step.

During optimization, the SOH estimator is exposed to causal hourly sequence segments, and benchmark execution processes one hourly aggregate at a time with causal hidden- and cell-state forwarding. This allows the estimator to accumulate degradation context without access to future information. The first hourly output is anchored with the configured initial SOH value before subsequent hourly predictions are taken directly from the recurrent model output. The SOH prediction is updated at hourly cadence and held piecewise constant between update points. This online update structure matches the gradually evolving capacity fade in the dataset and supplies the latest SOH context to the faster SOC estimators. In the HDM, the predicted SOH enters the SOC computation through the effective capacity relation  $C_{\text{eff}} = C_{\text{nom}} \cdot \widehat{\text{SOH}}$ , so the method remains structurally close to direct measurement while replacing the fixed-capacity assumption. In the HECM, the predicted SOH affects the capacity scaling and the lookup-based ECM parametrization, while in the DD model it is provided as an explicit SOC input feature that contextualizes the current measurement trajectory.

#### 2.4. Model and target-hardware preparation

The software benchmark uses trained floating-point model parameters and input scalers fixed before evaluation on the independent holdout test set. Neural training, hyperparameter selection, scaler fitting, pruning, and model selection use the declared training and validation sets. The HPPC-derived HECM lookup surfaces were identified from the same model-development sets and fixed without retuning before benchmark evaluation. Figure A.26 and Table A.4 document the complete cell-disjoint split. The compact SOC-GRU has hidden size 67 and the shared SOH-LSTM has hidden size 112. The continuous high-load test trajectory in Figure 6 is analyzed separately as a mechanism-focused lifecycle diagnostic.

The target-hardware experiment evaluates the isolated SOC cores on a NUCLEO-

246 H753ZI board with a 480 MHz STM32H753 Cortex-M7. To separate SOC imple-  
 247 mentation cost from the shared SOH service, the board receives the precomputed  
 248 causal SOH trace used by the software reference. The SOH-LSTM itself is not  
 249 executed on the microcontroller. Numerical equivalence and inference latency  
 250 are measured directly on the board. Execution time is recorded for every valid  
 251 inference with the processor-integrated hardware cycle counter, and each cell  
 252 sequence is replayed three times. Flash occupancy is calculated from the com-  
 253 piled firmware regions that must be stored in the microcontroller’s non-volatile  
 254 program memory.

255 Peak runtime RAM combines statically allocated variable storage, maximum  
 256 dynamic-memory allocation, and maximum temporary call-stack demand. The  
 257 static contribution is obtained from the compiled firmware, while dynamic allo-  
 258 cation is monitored during execution. Stack demand is measured with a high-  
 259 water-mark pattern during representative on-device replays that exercise the  
 260 complete estimator path. For DD, this includes valid inference after its rolling  
 261 input window has been filled. An unused compilation-time memory reserve is  
 262 excluded from the reported demand. Predictions are compared with both the  
 263 software implementation and dataset SOC. Figure 19 summarizes the resulting  
 264 memory requirements of the isolated SOC implementations with SOH supplied  
 265 as an external causal trace.

## 266 *2.5. Robustness benchmark design*

267 The primary cross-model benchmark manipulates current, voltage, tempera-  
 268 ture, sample availability, and sampling time, all of which are measured or recon-  
 269 structed online by a BMS. A separate initialization branch acts on deployment-  
 270 accessible estimator states. Parameter mismatch is excluded from this common  
 271 ranking and examined separately through the HECM lookup-table sensitivity.  
 272 Figure 3 groups the primary scenarios into input disturbances, initialization  
 273 errors, and signal-integrity faults, while Table 2 defines their implementation  
 274 [6].

275 The disturbance magnitudes follow three practical levels of justification:

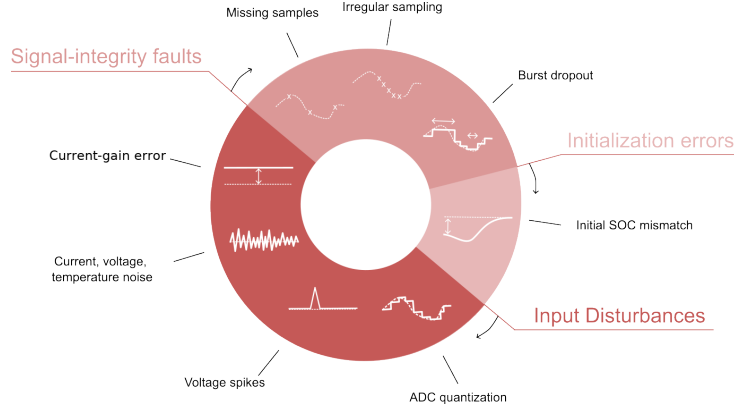


Figure 3: Taxonomy of the disturbance scenarios used in the performance benchmark, grouped into input disturbances, initialization errors, and signal-integrity faults.

276 deployment-level sensing limits, adverse measurement corruption, and signal-  
 277 integrity stress conditions [20, 21, 22, 23, 24, 25]. Input disturbances include  
 278 gain error, sensor noise, voltage spikes, and ADC quantization. The remaining  
 279 scenarios address estimator initialization, sample availability, and sampling-time  
 280 integrity. Together they form a controlled and reproducible benchmark matrix  
 281 for exposing estimator-specific failure mechanisms under a common protocol.

Table 2: Disturbance configurations used in the benchmark and their practical rationale. Exact numeric values are fixed benchmark settings. The cited sources motivate disturbance type and magnitude range.

| Scenario           | Benchmark setup                                                                                                                            | Practical origin / rationale                                                                 |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| ADC quantization   | $\Delta I = 0.01$ A,<br>$\Delta U = 0.005$ V, and<br>$\Delta T = 0.5$ °C                                                                   | ADC steps, sensor scaling, rounding, and coarse deployment-level discretization [20, 21, 26] |
| Current gain error | $I_{\text{meas}} = I(1 + b)$ with<br>$b = \pm 0.5\%$ , $\pm 1.5\%$ , and<br>$\pm 3.0\%$ . Both signs are<br>evaluated at each<br>magnitude | Sensor gain error, miscalibration, drift, and upper bound as stress case [21, 22, 23]        |

Table 2: Disturbance configurations used in the benchmark and practical rationale (continued).

| Scenario           | Benchmark setup                                                                                                           | Practical origin / rationale                                                                        |
|--------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Current noise      | Gaussian with $\sigma_I = 0.02$ A and 0.10 A                                                                              | Sensing-chain noise, EMI pickup, and low and high sensor-noise levels [4, 5]                        |
| Voltage noise      | Gaussian with $\sigma_U = 0.01$ V                                                                                         | Wiring noise, front-end noise, poor filtering, and EMI [4, 20, 24]                                  |
| Temperature noise  | Gaussian with $\sigma_T = 1.0$ °C                                                                                         | Sensor tolerance, conversion noise, thermal gradients, and sensor-to-cell mismatch [26, 11]         |
| Voltage offset     | Additive $U_{\text{meas}} = U + 0.02$ V                                                                                   | Front-end offset, calibration residual, wiring contribution, and adverse sensing stress             |
| Temperature offset | Additive $T_{\text{meas}} = T + 3$ °C                                                                                     | Sensor placement error, calibration residual, and cell-to-sensor thermal mismatch                   |
| Initial SOC error  | Initial mismatch of $-0.10$ SOC, with DD perturbed through online $Q_c$                                                   | Restart, storage, missing recent history, and weak LFP OCV anchoring [4, 8]                         |
| Missing samples    | Every 50th sample frozen and separate random 2% loss                                                                      | Packet loss, blocked updates, logger dropouts, and periodic and random data freeze [5, 25]          |
| Irregular sampling | Uniform jitter: $\pm 0.1$ s, $\pm 0.5$ s, and $\pm 0.9$ s around nominal 1 s grid                                         | Scheduling jitter, bus latency, and asynchronous acquisition [6, 5, 25]                             |
| Burst dropout      | One frozen 3600 s interval selected by maximum absolute net charge, with at least 12 h before and 24 h after the interval | Communication interruption, frozen task, stuck data stream, and measurement-only severe case [6, 5] |
| Voltage spikes     | Single-sample spikes of $\pm 0.20$ V every 1000 samples (1000 s)                                                          | EMI, switching disturbance, and corrupted voltage samples [24]                                      |

## 2.6. Metrics and evaluation criteria

For deployment-oriented interpretation, the benchmark results are synthesized along three performance dimensions: nominal accuracy under baseline conditions, recovery after initialization mismatch, and robustness under sustained or transient measurement corruption and signal unavailability. Disturbance-related global performance is quantified through MAE, RMSE, bias, maximum error, and the 95th percentile absolute error, because these metrics provide a compact view of average performance, systematic deviation, and tail behavior across the selected evaluation windows. Local disturbance performance is analyzed through scenario-specific metrics such as jump counts, output variance, absolute-error variance, and excess rolling MAE, because several disturbances do not primarily manifest as window-wide MAE collapse but as short-term output fluctuations, delayed drift, or slow re-synchronization. Whenever a disturbance mask exists, the analysis further separates pre-disturbance, disturbed, and post-disturbance behavior in order to quantify disturbance-window error growth and residual post-disturbance error.

ADC quantization, current-gain error, voltage and temperature disturbances, missing samples, and irregular sampling are interpreted primarily through global metrics. Burst dropout, voltage spikes, current-noise detail, and initialization mismatch additionally use dedicated local evidence because their most relevant effects are episodic. The canonical initialization-recovery analysis compares each perturbed estimator trajectory with its cell-, window-, model-, SOH-, and seed-matched correctly initialized trajectory. The resulting paired trajectory difference is the recovery signal, while absolute error against dataset SOC remains the accuracy measure. A first qualifying entry occurs when the paired difference enters a 0.02 SOC band for at least 300 s. Persistent recovery is the first subsequent point after which the difference remains within this band for the rest of the 24-h observation horizon. Runs without this endpoint are right-censored. First-entry time, later departure from the band, censored and relapse fractions, and excess-error area under the curve provide complementary recovery diagnostics. The protocol maps a  $-0.10$  SOC-equivalent mismatch into the

coulomb-counting SOC initialization for DM and HDM, the EKF SOC state for  
 HECM, and the capacity-equivalent  $Q_c$  offset for DD. Because DD first requires  
 a causal input context, recovery is evaluated for every estimator from the first  
 time at which all four provide a valid output. Recovery times remain refer-  
 enced to the original intervention. Endpoints already satisfied at the common  
 evaluation start are treated as left-censored and recorded as conservative up-  
 per bounds. The comparison therefore measures observable re-anchoring after  
 deployment-accessible, estimator-specific initialization interventions.

Each holdout cell is treated as one independent statistical unit. Protocol-  
 defined SOH windows and random seeds are first summarized within each cell,  
 after which the six cells receive equal macro weight. Hierarchical bootstrap  
 resampling with seeds nested inside cells and 10,000 repetitions provides 95%  
 confidence intervals. Scenario effects are evaluated as paired, cell-matched dif-  
 ferences from baseline, and estimator pairs are compared using exact two-sided  
 sign-flip tests, raw mean and median differences, paired standardized effect sizes,  
 and Holm-adjusted  $p$ -values. Holm correction is applied within each reported  
 scenario family. With six independent cells, effect magnitudes and confidence  
 intervals provide the primary evidence and  $p$ -values support their interpretation.  
 The full machine-readable tests accompany the result tables, while the baseline  
 pair tests are reproduced in the Appendix.

To ensure like-for-like comparisons, all global accuracy, robustness, stratified,  
 confidence-interval, and paired-test calculations begin when the rolling DD first  
 provides a valid estimate. The resulting common evaluation mask gives every  
 estimator the same source interval and the same number of scored observations.

The decision synthesis provides a relative summary of the four implemen-  
 tations. Accuracy combines baseline MAE, RMSE, and 95th-percentile error.  
 Robustness assigns equal weight to eight evaluated disturbance families: sensor  
 noise, current-gain error, sensor offsets, quantization, missing samples, timing  
 jitter, burst dropout, and voltage spikes. The current-gain family uses the ad-  
 verse direction from the matched signed sweep. Non-positive  $\Delta$ MAE values con-  
 tribute zero disturbance penalty. Levels within a family are averaged before the

family scores are combined, giving every disturbance family equal influence. Recovery combines observed persistent recovery-or-censor time, excess-error area, and persistent censored fraction from the canonical paired initialization analysis. Relapse after a first qualifying entry remains a separate diagnostic because it is defined only for runs that enter the recovery band. Equal-scenario and high-severity weighting variants quantify the sensitivity of the summary to alternative deployment priorities.

### 3. Experimental Setup

#### 3.1. Dataset and data acquisition

The benchmark uses the MGFarm LFP 18650 dataset because its controlled design-of-experiments aging study provides long 1-Hz full-cell trajectories, repeated capacity checkups, multiple operating conditions, and enough independent cells for a cell-disjoint comparison [27, 28, 29]. The fixed split comprises a six-cell training set, a three-cell validation set, and a six-cell independent holdout test set. Figure A.26 shows the complete SOH trajectories together with their charge-rate, discharge-rate, and depth-of-discharge aging conditions. Table A.4 provides the exact cell-to-set assignment and reports the measured duration, SOH, temperature, and load coverage. Figure A.21 summarizes the diversity within the holdout test set. Together, these materials document the cell-disjoint development boundary and the operating diversity of the evaluation data. LFP’s flat mid-SOC OCV and chemistry-specific aging behavior are integral to the tested problem.

The holdout test set is assigned to descriptive load groups using the measured 95th-percentile absolute C-rate. It contains two low-load cells, three middle-load cells, and one high-load cell. The single high-load trajectory is exploratory, whereas the low- and middle-load groups support multi-cell summaries. The exact assignment and the corresponding load coverage are shown in Figure A.21 and Table A.4. These cell-level load groups describe the operating coverage of the holdout set and are distinct from the multivariate selection of representa-

373 tive evaluation windows. The latter uses measured SOH, temperature, C-rate,  
 374 charge-throughput, and SOC-coverage descriptors as detailed in Section 3.3.  
 375 This separation preserves variation across cells and aging states without treat-  
 376 ing every operating-state bin as an independent statistical sample.

### 377 3.2. Dataset ground truth and preprocessing

378 The reference preprocessing first computes the signed transferred charge as  
 379  $Q_k = I_k \Delta t_k / 3600$  and the absolute transferred charge as  $|Q_k|$ , from which the  
 380 capacity during dedicated capacity-test intervals is obtained as  $C_{\text{ref}} = \sum_k |Q_k|$ .  
 381 The reference SOH target is then defined offline as  $\text{SOH}_{\text{ref}} = C_{\text{ref}} / C_{\text{ref},0}$ , where  
 382  $C_{\text{ref},0}$  denotes the first measured reference capacity of the trajectory, and the  
 383 resulting capacity and SOH columns are interpolated between the dedicated  
 384 capacity-test anchors. The dataset SOC ground truth used in the benchmark  
 385 is reconstructed offline as  $\text{SOC}_{\text{ref}} = 1 + Q_c / C_{\text{ref},0}$ , where the cumulative charge  
 386 state  $Q_c$  is propagated from the signed charge and reset to 0 at the upper  
 387 voltage anchor and to  $-C_{\text{ref},0}$  at the lower voltage anchor during preprocessing.  
 388 This reconstructed SOC serves as the common dataset ground truth for every  
 389 reported SOC error. It is an offline evaluation quantity because its dataset-side  
 390 processing and anchor logic are not fully available to an online estimator during  
 391 deployment. All nominal accuracy values therefore refer to this consistently  
 392 applied dataset ground truth. Paired  $\Delta\text{MAE}$  compares each disturbed run with  
 393 its matched baseline under the same ground truth.

### 394 3.3. Test trajectory and benchmark scenarios

395 All estimators are evaluated on every available cell/SOH combination in the  
 396 holdout test set under the same disturbance protocol. One representative 24-  
 397 h window is frozen for each available fresh, mid-life, and aged state, yielding  
 398 16 windows. One test trajectory remains entirely within the fresh SOH range  
 399 and therefore contributes no mid-life or aged window, as shown in Figure A.21.  
 400 Every candidate window begins at the same measured full-charge condition  
 401 ( $\text{SOC} \geq 0.98$ ,  $U \geq 3.58$  V). Multiple eligible windows can nevertheless exist for

one cell and SOH state, and their temperature, load, charge throughput, and SOC coverage can differ substantially. Each candidate window  $w$  is therefore represented by the measured operating-condition vector

$$\mathbf{z}_w = \left[ \widetilde{\text{SOH}}_w \quad \bar{T}_w \quad P_{95}(T)_w \quad P_{95}(|C_{\text{rate}}|)_w \quad Q_{\text{through},w} \quad f_{\text{SOC} < 0.2,w} \right]^\top, \quad (9)$$

where the entries denote median SOH, mean temperature, 95th-percentile temperature, 95th-percentile absolute C-rate, absolute charge throughput, and the fraction of samples below 20% SOC. For every descriptor  $j$ , the candidate values are centered by their median  $m_j$  and scaled by their median absolute deviation  $s_j$ . The standard deviation is used when the median absolute deviation is zero, and an invariant descriptor receives unit scale. The selected observed window is

$$w^* = \arg \min_w \sqrt{\sum_j \left( \frac{z_{w,j} - m_j}{s_j} \right)^2}. \quad (10)$$

This measured-feature medoid criterion selects the actual window closest to the typical multivariate operating profile within each cell and SOH state. It avoids arbitrary or estimator-favorable window selection because neither model predictions nor errors enter the criterion. Applying the same procedure separately to every holdout cell and available aging state retains diversity across cells and SOH conditions while preventing unusual operating periods from dominating the comparison.

The one-hour dropout uses 48 h from the same full-charge anchor to retain approximately 24 h of post-gap observation. The shared SOH LSTM receives 192 h of undisturbed causal context before every scored window, matching its trained sequence length. Figure A.23 summarizes the selection and evaluation protocol.

The benchmark contains 19 scenario definitions: the nominal baseline, two current-noise levels, voltage and temperature noise, three signed current-gain magnitudes, voltage and temperature offsets, ADC quantization, initial SOC mismatch, periodic and random sample loss, three timing-jitter levels, a one-hour burst dropout, and randomized-sign voltage spikes. Each current-gain

429 magnitude contains matched positive and negative sign sublevels. Adverse-  
 430 direction summaries report the larger paired  $\Delta\text{MAE}$  of the two signs at each  
 431 magnitude, capturing the larger sensor-gain penalty when signed compensation  
 432 with baseline error occurs. Gaussian sensor-noise cases use 10 deterministic  
 433 seeds. Random sample loss, jitter, and spike cases use 5 seeds. Deterministic  
 434 cases run once. Figure A.22 maps the scenario definitions to the manipulated  
 435 measurement or state and marks the selected paired reference-SOH ablations.  
 436 Input disturbances act directly on measured channels, initialization interven-  
 437 tions act on each estimator’s deployment-accessible state, and signal-integrity  
 438 scenarios alter sample availability or timing while preserving causal execution.

### 439 *3.4. Implementation details*

440 Every estimator is executed causally, and all auxiliary quantities required  
 441 by the models are rebuilt directly from disturbed inputs. This applies to  $Q_c$ ,  
 442 EFC,  $dU/dt$ ,  $dI/dt$ , and hourly SOH aggregates, ensuring that estimator inputs  
 443 originate from the causal measurement path. The SOH LSTM forwards hidden  
 444 and cell states across hourly updates after its 192-h initialization context. The  
 445 primary SOC-GRU evaluation preserves its trained sequence-to-one semantics:  
 446 each prediction uses the latest 2024-sample rolling window with recurrent state  
 447 reset between windows. The continuous-stateful stream forms a separate de-  
 448 ployment configuration whose full-trajectory accuracy and resource demand are  
 449 reported alongside the rolling implementation. In freeze and dropout scenar-  
 450 ios, disturbed or frozen measurements propagate consistently into derived online  
 451 features. The dropout start is selected from measured current as the eligible one-  
 452 hour interval with the largest absolute integrated net charge. Selection enforces  
 453 at least 12 h of pre-gap observations and 24 h of post-gap observations and re-  
 454 mains independent of dataset SOC and estimator output. Direct-measurement  
 455 estimators use the same CV-triggered reset that defines the full-charge start,  
 456 whereas the DD initial-SOC test uses an equivalent offset of online  $Q_c$  because  
 457 the network exposes no explicit SOC state. The environment enforces common  
 458 nominal-capacity metadata, disturbed streams, targets, masks, and metrics.

459 Model-specific capacity handling remains part of the estimator definitions.

### 460 3.5. HECM lookup-table sensitivity

461 A separate HECM-only analysis quantifies whether its current-gain response  
462 depends on exact lookup-table calibration. It applies one systematic perturba-  
463 tion at a time to the fixed charge and discharge surfaces. The  $R_i$ ,  $R_1$ , and  
464  $R_2$  surfaces are jointly scaled by  $-10\%$  or  $+10\%$ , while the OCV surfaces are  
465 shifted by  $-10$  mV or  $+10$  mV. The OCV derivative and time-constant surfaces  
466 remain fixed. Each perturbation and the nominal lookup are evaluated under  
467 baseline current and matched  $-3\%$  and  $+3\%$  current-gain errors across all frozen  
468 windows. Results are first averaged within each cell and then aggregated using  
469 the same equal-weight cell-bootstrap procedure as the main benchmark. For  
470 each lookup condition, the adverse gain penalty is the larger cell-level  $\Delta\text{MAE}$   
471 from the two gain signs. The lookup-gain interaction subtracts the nominal-  
472 lookup adverse penalty from the perturbed-lookup adverse penalty. These one-  
473 at-a-time shifts are controlled local sensitivity probes rather than probabilistic  
474 uncertainty bounds. They remain outside the cross-model robustness score.

## 475 4. Results

### 476 4.1. Baseline performance

477 Figure 4 summarizes nominal accuracy against the dataset ground truth  
478 across the 16 predefined windows on the common evaluation interval. Equal  
479 weighting within the six-cell holdout test set gives the lowest baseline MAE to  
480 DD at 0.0258 (95% CI 0.0189–0.0325), followed by HECM at 0.0337 (0.0246–  
481 0.0434), HDM at 0.0412 (0.0301–0.0515), and DM at 0.0690 (0.0499–0.0823).  
482 The RMSE ranking is identical, with values of 0.0300, 0.0388, 0.0442, and  
483 0.0740, respectively. Cell-level points reveal substantial between-cell variation,  
484 particularly for DM, and place the cell-macro means in the context of heteroge-  
485 neous trajectories. Exact paired sign-flip tests support the magnitude and direc-  
486 tion of the principal contrasts. With six independent cells, the Holm-adjusted

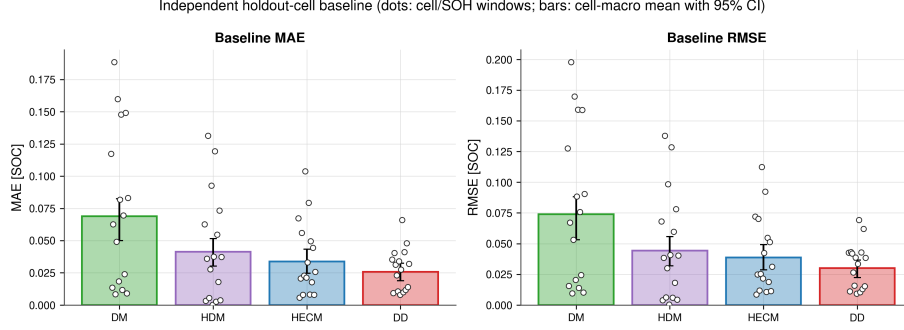


Figure 4: Baseline accuracy against the dataset SOC ground truth across the independent holdout test set. Dots show cell/SOH-window values. Bars and error bars show the equal-weight cell-macro mean and hierarchical 95% confidence interval. DD has the lowest macro MAE and RMSE, while DM has the largest error and between-cell spread.

values remain above 0.05, so confidence intervals and paired effect sizes provide the main statistical evidence reported in the Appendix. These values quantify accuracy against the consistently reconstructed dataset ground truth.

#### 4.2. Current-gain sensitivity

Figure 5 summarizes the matched signed current-gain sweep at  $\pm 0.5\%$ ,  $\pm 1.5\%$ , and  $\pm 3.0\%$ . For each magnitude and holdout cell, the adverse-direction summary uses the larger MAE change from the two signs because one direction can partially compensate a pre-existing baseline error. The resulting degradation increases monotonically with gain-error magnitude for every estimator. DM and HDM show the strongest response, HECM is less sensitive, and DD is least sensitive. At 3%, the cell-macro MAE increases by 0.0108 for DM, 0.0101 for HDM, 0.0060 for HECM, and 0.0038 for DD. The complete signed sweep remains available for interpreting direction-specific behavior.

The HECM lookup-table analysis in Figure A.25 separates baseline calibration effects from current-gain robustness. Across the tested resistance and OCV perturbations, baseline MAE ranges from 0.0314 to 0.0373, whereas the adverse 3% current-gain penalty remains within a much narrower range of 0.00580 to 0.00628. Relative to the nominal lookup, the largest absolute macro interaction

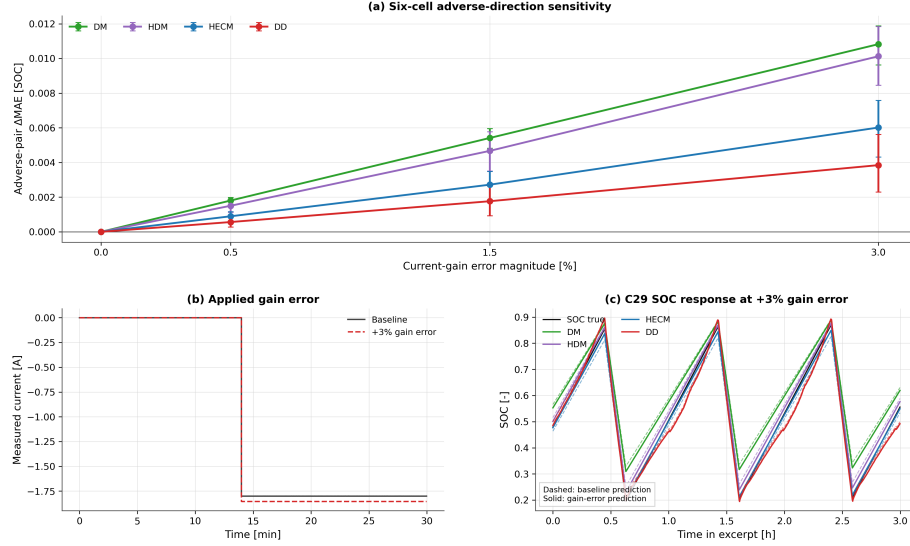


Figure 5: Current-gain sensitivity. (a) Six-cell adverse-direction  $\Delta\text{MAE}$  for each matched  $\pm b$  pair with hierarchical 95% confidence intervals. (b) Representative current before and after the +3% gain error. (c) SOC trajectories from the representative high-load test cell over three charge–discharge intervals. The test-set load assignment is documented in Figure A.21. The adverse-pair summary reports the larger penalty from each matched sign pair.

505 is 0.00026 MAE, and every 95% interval includes zero. Within these local per-  
 506 turbation ranges, lookup calibration therefore affects absolute HECM accuracy  
 507 more than its incremental current-gain penalty.

508 The continuous high-load test-set analysis in Figure 6 complements the  
 509 window-based benchmark and separates three mechanisms that independent  
 510 24-h windows cannot resolve. The gain-error contribution varies over life be-  
 511 cause its signed Coulomb-counting error interacts with charge direction and  
 512 operating state. Between full charges, the penalty often grows within an inter-  
 513 val, but full-charge anchoring can partially reduce or restructure it. The process  
 514 is therefore neither an unrestricted monotonic drift nor a complete reset. Across  
 515 this single full-life replay, the adverse 3% gain error increases MAE by 0.0102 for  
 516 DM, 0.0072 for HDM, 0.0037 for HECM, and 0.0048 for DD. HECM therefore  
 517 has the smallest full-life gain-error contribution in this trajectory, while DM and

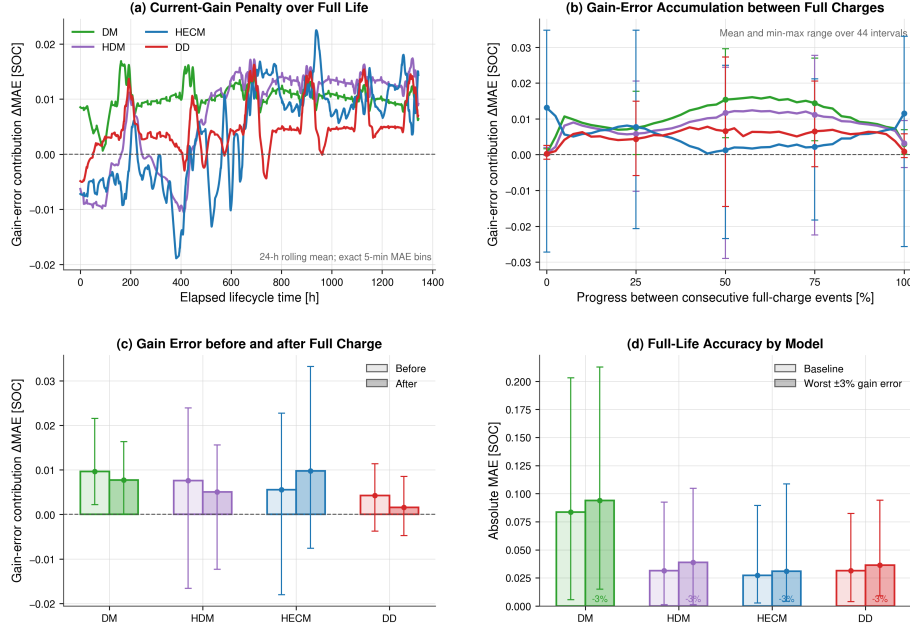


Figure 6: Current-gain-error diagnostic over the continuous high-load holdout trajectory. (a) Rolling 24-h gain-error contribution over elapsed life. (b) Mean and range over normalized intervals between consecutive full-charge events. (c) Penalty immediately before and after full-charge events. (d) Full-life baseline and adverse-gain MAE. Full-charge events reduce or restructure accumulated error but do not erase the lifetime penalty uniformly.

518 HDM remain most exposed. This mechanism-focused diagnostic is not a six-cell  
 519 lifecycle ranking.

#### 520 4.3. Noise sensitivity

521 Repeated sensor noise produces smaller global penalties than persistent cur-  
 522 rent gain error, with a response that depends on the disturbed channel and  
 523 estimator structure. At  $\sigma_I = 0.10$  A, HECM has the largest mean macro  
 524 penalty ( $\Delta\text{MAE} = 0.0038$ ), although its interval spans zero. DM changes by  
 525  $-0.0009$ , while HDM and DD remain near zero. These small signed changes  
 526 arise from finite-window compensation around imperfect baseline trajectories.

527 Voltage noise ( $\sigma_U = 0.01$  V) produces a 0.0014 penalty for DM and 0.0013 for  
 528 HDM, while HECM and DD remain below 0.0003. Temperature noise is neutral  
 529 for DM and small for the SOH-aware models. Figure 7 shows the local trans-  
 530 mission of current noise, while Figure 8 reports repeated six-cell effects with  
 531 hierarchical uncertainty.

532 The offset cases expose a different pattern from zero-mean noise. A  $+3^\circ\text{C}$   
 533 temperature offset increases DD MAE by 0.0074 (95% CI 0.0027–0.0125), while  
 534 the corresponding changes for DM, HDM, and HECM are 0.0000, 0.0002, and  
 535 0.0002. A  $+0.02$  V voltage offset increases DD by 0.0038 and HDM by 0.0005,  
 536 leaves DM effectively unchanged, and changes HECM by  $-0.0040$  in these finite  
 537 windows. The signed HECM result reflects compensation relative to its baseline  
 538 trajectory. These offset results are included in the cross-scenario heatmap and  
 539 in the family-balanced robustness synthesis.

#### 540 4.4. Initialization error and recovery

541 Initialization performance uses the paired criterion defined above. The pro-  
 542 tocol maps the same  $-0.10$  SOC-equivalent mismatch to the explicit DM/HDM  
 543 Coulomb-counting state, the HECM EKF SOC state, and a  $Q_c$  offset of  $-0.10C_{\text{nom}}$   
 544 for DD. Figure 9 reports the resulting paired trajectory differences from the  
 545 common evaluation start, while recovery time remains referenced to the inter-  
 546 vention. HECM provides the fastest persistent recovery, with a cell-macro mean  
 547 of 1.20 h, followed by DD at 2.60 h. DD enters the recovery band temporarily in  
 548 several runs but has a higher relapse fraction than HECM (0.78 compared with  
 549 0.17). DM and HDM generally do not recover persistently within the observa-  
 550 tion horizon, and both have a persistent-censor fraction of 0.89. HECM also  
 551 has the smallest excess-error area at 0.078 SOC h, compared with 0.116 SOC h  
 552 for DD. Together, the recovery time, relapse behavior, censoring, and excess-  
 553 error area identify HECM as the strongest recovery implementation under this  
 554 protocol.

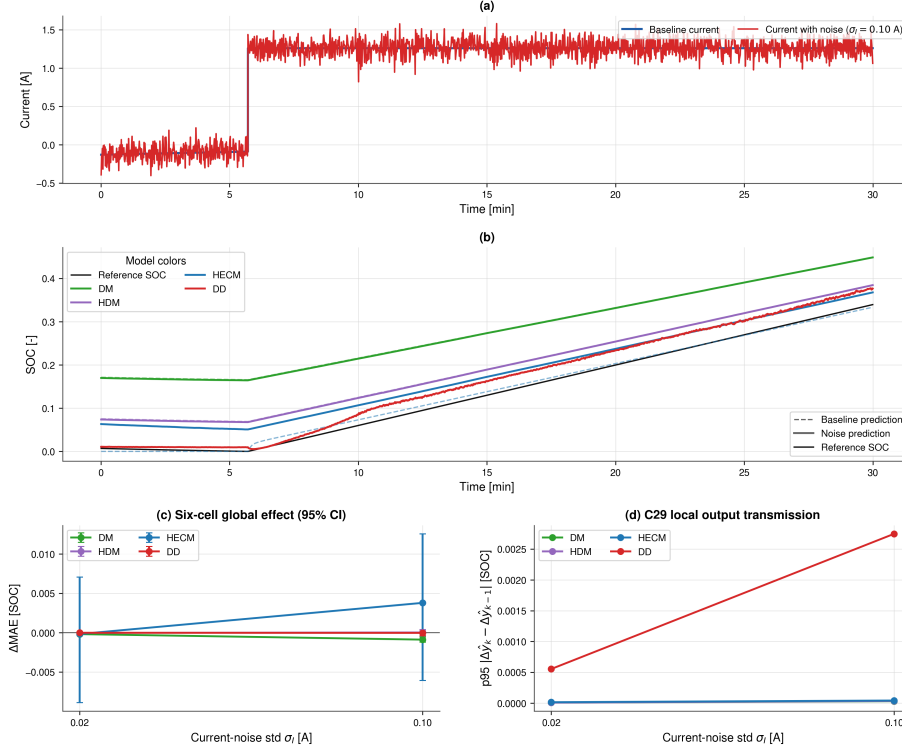


Figure 7: Noise sensitivity under additive current noise. (a) Baseline and noisy current in a reset-free 30 min example from the high-load test cell. (b) Matched baseline and noisy SOC trajectories in the same window. (c) Cell-macro  $\Delta\text{MAE}$  with hierarchical 95% confidence intervals across the holdout test set. (d) Local transmission in the same example, measured as the 95th percentile of the change in noise-induced output deviation between consecutive samples. The local panels illustrate the mechanism, while panel (c) provides the test-set comparison.

#### 4.5. Missing samples and signal integrity

Figure 10 compares the six-cell effects of periodic and random sample loss with those of timing jitter. Sample loss is substantially more consequential than timing jitter. Periodic freezing of every 50th sample increases MAE by 0.0072, 0.0065, 0.0028, and 0.0011 for DM, HDM, HECM, and DD, respectively. Random 2% loss produces corresponding increases of 0.0080, 0.0081, 0.0039, and 0.0026. By contrast, even the  $\pm 0.9$  s jitter case remains within approximately

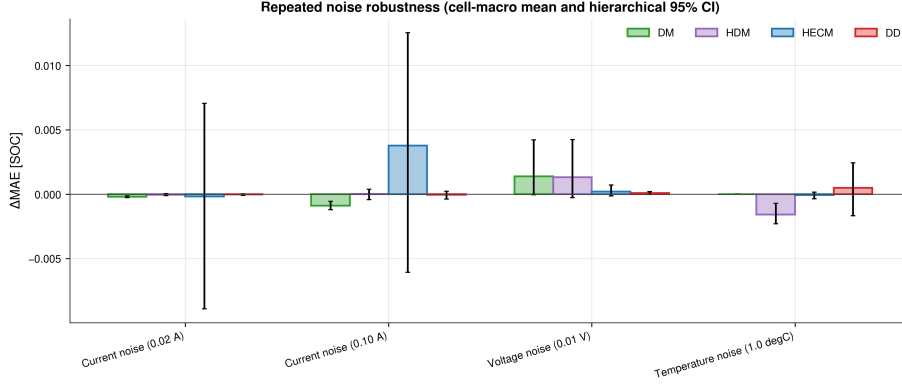


Figure 8: Repeated noise robustness across the six-cell holdout test set. Bars show cell-macro  $\Delta\text{MAE}$  and error bars show hierarchical 95% confidence intervals with random seeds nested within cells. HECM is most affected by high current noise, while most other channel-model combinations remain close to baseline relative to the between-cell uncertainty.

562  $6 \times 10^{-4}$  SOC of baseline for every model, with confidence intervals generally  
 563 spanning zero. DD therefore has the smallest sample-loss penalty, while DM and  
 564 HDM retain their structural dependence on uninterrupted charge integration.

565 The one-hour burst dropout creates a distinct availability problem. Against  
 566 the duration-matched 48-h baseline, its macro  $\Delta\text{MAE}$  is largest for HECM  
 567 (0.0239, 95% CI 0.0155–0.0366), followed by HDM (0.0209, 0.0098–0.0313) and  
 568 DM (0.0050,  $-0.0029$ –0.0138). DD changes by  $-0.0093$  ( $-0.0288$ –0.0099), re-  
 569 flecting finite-window compensation between its baseline and frozen-input tra-  
 570 jectories. The interval is placed where the absolute unobserved net charge is  
 571 largest among starts that retain the required pre- and post-gap context. Fig-  
 572 ure 11 compares the disturbed output of the representative high-load test cell  
 573 with its duration-matched no-dropout trajectory and reports the global holdout-  
 574 test effect. The paired local deviation has model-specific origins: DM/HDM  
 575 retain the unobserved charge deficit until a physical anchor, HECM corrects  
 576 through weak LFP voltage observability, and DD changes as its rolling input  
 577 context is replaced by valid samples. Dropout contributes to the robustness  
 578 dimension, while the recovery dimension uses the canonical paired initialization

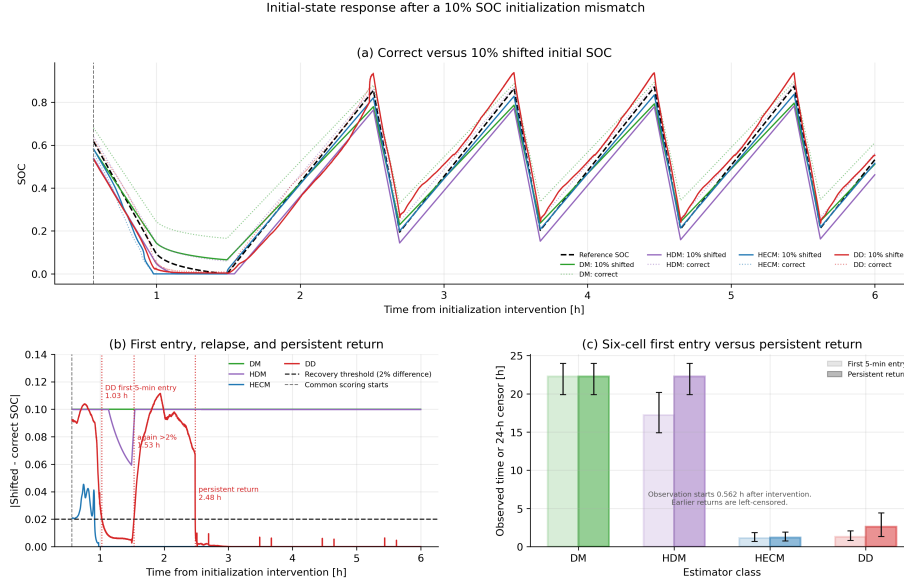


Figure 9: Response to a 10% SOC-equivalent initialization mismatch. (a) Correctly and incorrectly initialized trajectories for a representative mid-life pair from the high-load holdout trajectory. (b) Absolute paired differences over the first 6 h. The DD trajectory enters the 2% band for 5 min, leaves it again, and only later returns persistently. The vertical gray marker denotes the common evaluation start after all estimators provide valid output. (c) Observed first qualifying entry and persistent recovery-or-censor time across the holdout test set with hierarchical 95% confidence intervals. Endpoints already satisfied at the common start are left-censored.

579 experiment.

#### 580 4.6. Spike sensitivity

581 Single voltage spikes demonstrate why local and global metrics must be re-  
582 ported together. Full-window  $\Delta\text{MAE}$  remains below  $4 \times 10^{-4}$  SOC for every  
583 estimator and is effectively zero for DM, HDM, and DD. Event-aligned analysis,  
584 however, identifies a reproducible DD transient: its mean spike-sample penalty  
585 across cells is approximately  $7.7 \times 10^{-3}$  SOC, compared with near-zero direct  
586 penalties for the other representatives. Figures 12 and 13 further show that the  
587 DD response varies across cells and SOH states and then decays after the event.  
588 This pattern is consistent with a one-sample voltage impulse entering both the

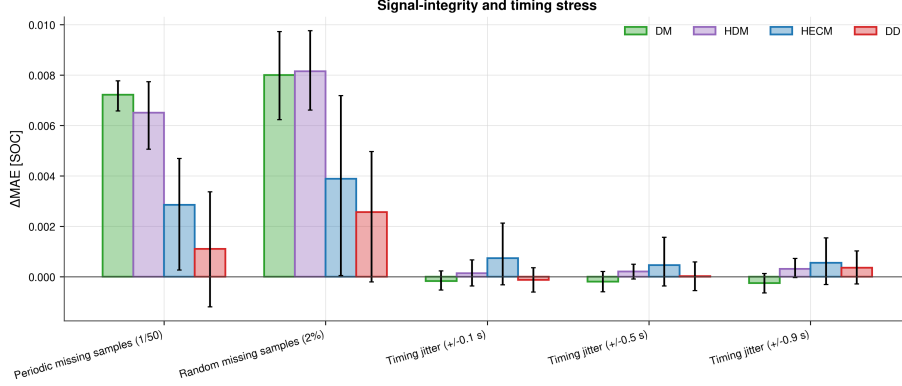


Figure 10: Six-cell signal-integrity and timing stress. Bars show cell-macro  $\Delta\text{MAE}$  and hierarchical 95% confidence intervals for periodic and random sample loss and three timing-jitter levels. Sample loss clearly dominates timing jitter under the tested protocol.

589 absolute voltage and  $dU/dt$  channels and perturbing the rolling recurrent pre-  
 590 diction. A channel-removal ablation would separate the individual contributions  
 591 of the absolute and derivative inputs.

#### 592 4.7. ADC quantization

593 Figure 14 compares representative raw and quantized current and voltage  
 594 signals and summarizes the resulting six-cell  $\Delta\text{MAE}$ . The tested ADC steps  
 595 ( $\Delta I = 0.01$  A,  $\Delta U = 0.005$  V, and  $\Delta T = 0.5$  °C) preserve the local signal shape  
 596 but affect the tested representatives differently. DM and HDM increase by  
 597 0.0038 and 0.0035 MAE, respectively. HECM changes by  $-0.0018$  and DD by  
 598  $+0.0002$ . All four confidence intervals overlap zero. At the selected coarse reso-  
 599 lution, quantization produces smaller penalties than current gain error, sample  
 600 loss, and burst dropout, while the integration-based representatives retain a  
 601 measurable response.

#### 602 4.8. Shared-SOH ablation

603 The primary comparison supplies one common causal LSTM-SOH trace to  
 604 HDM, HECM, and DD. A paired ablation repeats selected scenarios with dataset  
 605 reference SOH to isolate the influence of this shared service. Reference SOH

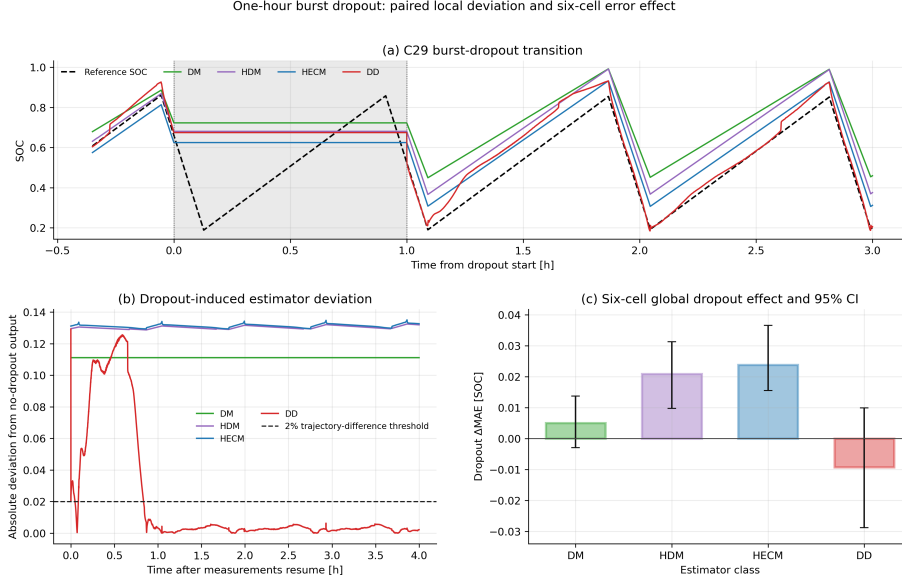


Figure 11: One-hour burst-dropout transition. (a) Estimator outputs around the shaded missing-input interval. (b) Absolute deviation from the paired duration-matched no-dropout output after measurements resume, with the 2% trajectory-difference line shown for context. (c) Six-cell global  $\Delta\text{MAE}$  with hierarchical 95% confidence intervals.

606 reduces baseline MAE by 0.0370 for HDM and 0.0166 for HECM, while DD  
607 MAE increases by 0.0049 on average. The same direction persists for most  
608 noise, sample-loss, and dropout cases. At 3% current gain error, reference SOH  
609 improves HDM and HECM and changes DD by +0.0091. For temperature noise,  
610 the reference-minus-LSTM difference changes from its baseline value by 0.0016  
611 for HDM and less than  $10^{-4}$  for HECM and DD, indicating that the incremental  
612 temperature-noise response is largely independent of the shared SOH service.  
613 The reversed DD response reflects calibration and feature interaction with its  
614 SOH input. The primary LSTM-SOH results represent the complete causal  
615 estimator pipelines, while the reference-SOH runs explain how each SOC branch  
616 consumes health information.

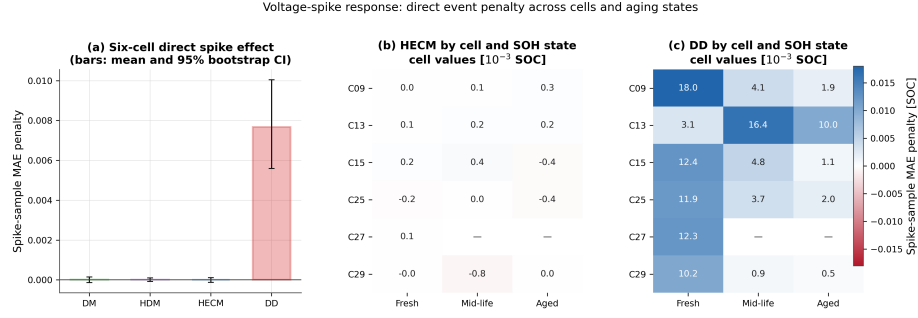


Figure 12: Voltage-spike sensitivity across six cells. (a) Mean event-sample MAE penalty with hierarchical 95% confidence intervals. (b,c) Cell- and SOH-state-resolved penalties for HECM and DD, respectively. Absent cell/state combinations are left blank. The event-aligned metric exposes the local DD response that is obscured in full-window  $\Delta$ MAE.

#### 4.9. Robustness synthesis across representatives

Figure 15 condenses 17 measurement-corruption and signal-integrity rows and shows how degraded-input behavior changes the nominal-accuracy picture. Persistent gain error and missing information create large global penalties. Current-gain sensitivity increases with magnitude for every representative in the matched signed sweep, periodic and random sample loss particularly affect DM and HDM, and the one-hour dropout produces the largest point penalty for HECM. Timing jitter remains nearly neutral. Voltage spikes reveal a distinct DD transient through event alignment despite their small full-window MAE. The alternative score aggregations then quantify how different priorities change the combined robustness order. Together, the heatmap and sensitivity analysis provide the scenario-specific evidence required for estimator selection.

Under the family-balanced definition, the robustness scores are 0.562 for DM, 0.468 for HDM, 0.428 for HECM, and 0.842 for DD. Equal weighting per evaluated scenario yields 0.574, 0.504, 0.560, and 0.781, while the highest-level-plus-offset variant yields 0.615, 0.512, 0.576, and 0.665. DD leads all three aggregations, while the changing score distances and HDM–HECM order demonstrate the influence of deployment priorities. The individual scenario results identify the disturbance mechanisms behind these aggregate changes.

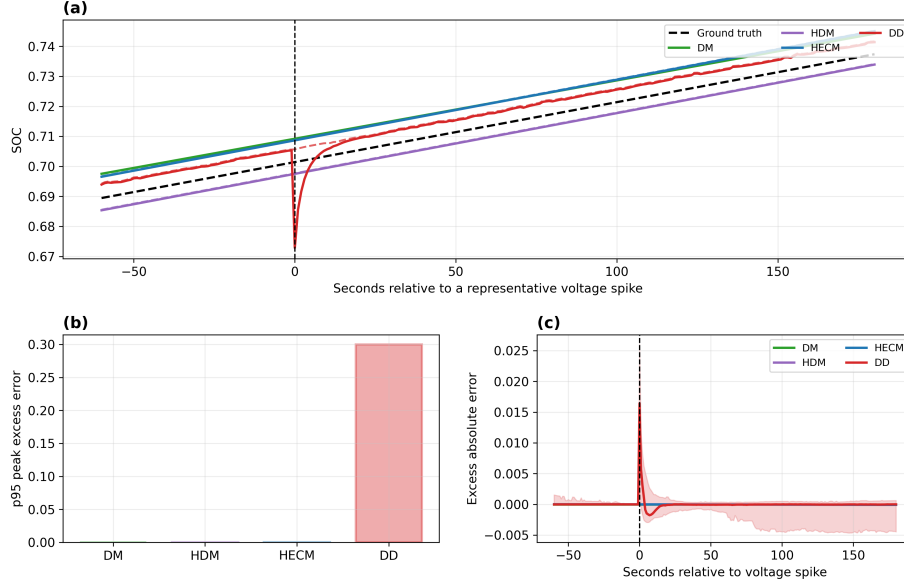


Figure 13: Event-aligned voltage-spike response. (a) Representative SOC trajectories around the injected spike, with the dashed black line showing dataset ground-truth SOC. (b) Model-wise 95th percentile peak excess error. (c) Mean excess absolute error aligned across repeated spikes, with the DD uncertainty envelope. The local response decays rapidly and is therefore weakly represented by full-window MAE.

#### 4.10. Embedded performance benchmark

The STM32 experiment first verifies numerical equivalence before comparing speed or memory. Across the six nominal cell sequences, mean hardware-software MAE remains below  $4 \times 10^{-4}$  percentage points for every SOC core (Figure 17). These differences are orders of magnitude below the estimator-to-dataset errors and show that the float32 C implementations reproduce their software references sufficiently closely for deployment measurements.

The isolated DM and HDM SOC updates both have sub-microsecond median latency (0.171 and 0.165  $\mu\text{s}$ ), HECM requires 23.9  $\mu\text{s}$ , and continuous-state DD requires 424.9  $\mu\text{s}$  (Figure 18). At the nominal 1-Hz update rate, these medians occupy less than 0.043% of one sampling period. This deadline share describes the isolated estimator task and provides the execution margin available for inte-

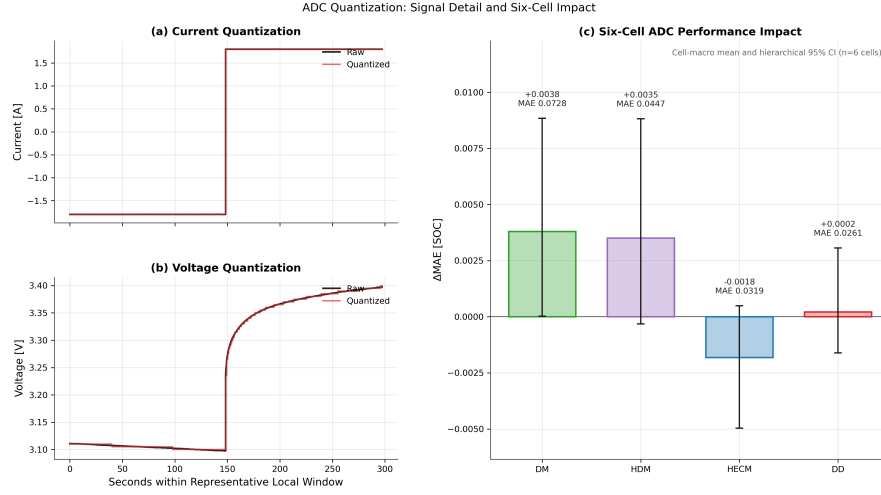


Figure 14: ADC quantization. (a,b) Representative raw and quantized current and voltage. (c) Six-cell  $\Delta\text{MAE}$  with hierarchical 95% confidence intervals and resulting quantized-signal MAE annotations.

648 gration with other BMS functions. The DD value in the cross-model comparison  
 649 is the continuous-state deployment variant. Exact 2024-sample rolling-window  
 650 semantics are evaluated separately because they recompute the full sequence at  
 651 every sample.

652 For the cross-model embedded performance comparison, DD denotes the  
 653 continuous-state deployment variant used in the latency analysis. The compiled  
 654 DM, HDM, HECM, and DD firmware occupies 27.0, 27.1, 470.1, and 115.9 KiB  
 655 of flash, respectively. The corresponding measured peak runtime RAM is 2.4,  
 656 2.4, 2.5, and 4.1 KiB (Figure 19). These peak values combine static storage, dy-  
 657 namic allocation, and call-stack demand. HECM flash occupancy is dominated  
 658 by its parameter surfaces, whereas the DD parameter storage accounts for most  
 659 of its flash demand. The measurements describe the isolated SOC firmware  
 660 with the shared SOH trace provided externally and leave resource margin for  
 661 additional BMS tasks.

662 Figure 20 makes the DD deployment trade-off explicit, while Figure A.24  
 663 provides the corresponding latency distributions over all six hardware cell se-

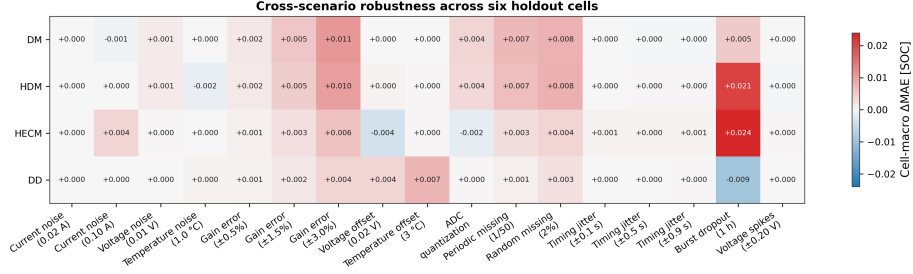


Figure 15: Cell-macro  $\Delta$ MAE across 17 measurement-corruption and signal-integrity rows on the six-cell holdout test set. Current-gain rows use the adverse direction from the matched signed sweep. Positive values denote degradation relative to each model’s paired baseline, while small negative values arise from finite-window signed compensation. The initialization-mismatch response is evaluated separately in Figure 9.

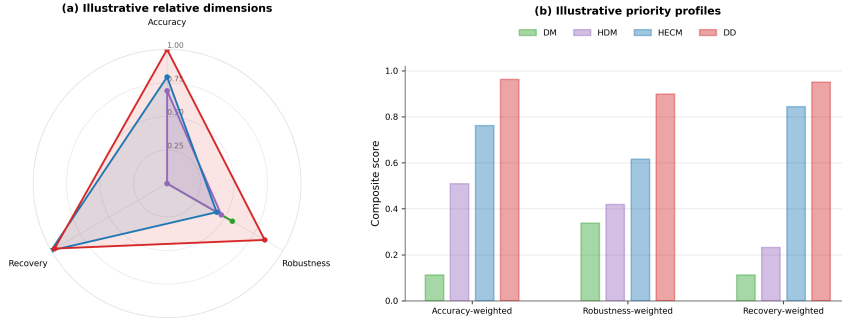


Figure 16: Decision-oriented synthesis for the four tested representatives. (a) Relative min-max-normalized accuracy, family-balanced robustness across eight evaluated disturbance families, and observed persistent paired recovery. The current-gain component uses the adverse direction from the matched signed sweep. (b) Composite scores under three 60/20/20 priority profiles. The profiles are interpreted together with the scenario-level evidence in Figure 15.

664 quences. Exact rolling-window inference preserves the trained software seman-  
665 tics but requires roughly 724 ms median inference time, or 72.4% of the one-  
666 second period, and 67.3 KiB peak runtime RAM. Continuous-state inference  
667 reduces latency to approximately 0.425 ms and peak RAM to 4.1 KiB while  
668 retaining similar error on the six nominal sequences. Periodically resetting that

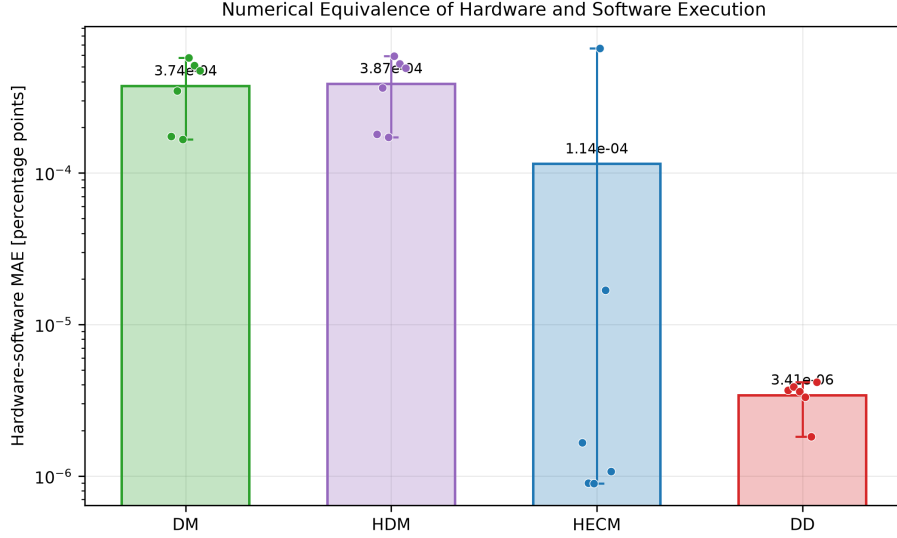


Figure 17: Numerical equivalence of STM32 and software execution over six cell sequences. Bars show mean hardware–software MAE in percentage points, dots show cells, and whiskers show the cell range. The logarithmic scale emphasizes that all implementation differences are negligible relative to model error.

state has a comparable 4.1 KiB peak but creates much larger transient errors, with a maximum SOC error near 28 percentage points. Continuous-state execution therefore provides the practical hardware configuration. Its forwarded recurrent state defines a deployment mode that is evaluated separately from the primary rolling-window benchmark.

## 5. Discussion

### 5.1. Structural interpretation of the representatives

The four implementations expose distinct relationships between their update mechanisms and observed disturbance responses. DM has the largest nominal error and reacts strongly to persistent gain error and missing samples because its current-integration structure lacks voltage-residual correction. Its practical full-charge anchor limits accumulated drift, while the lifecycle experiment shows that the reset restructures the gain-error contribution without removing

On-Device Inference-Latency Distributions Across Six Cells

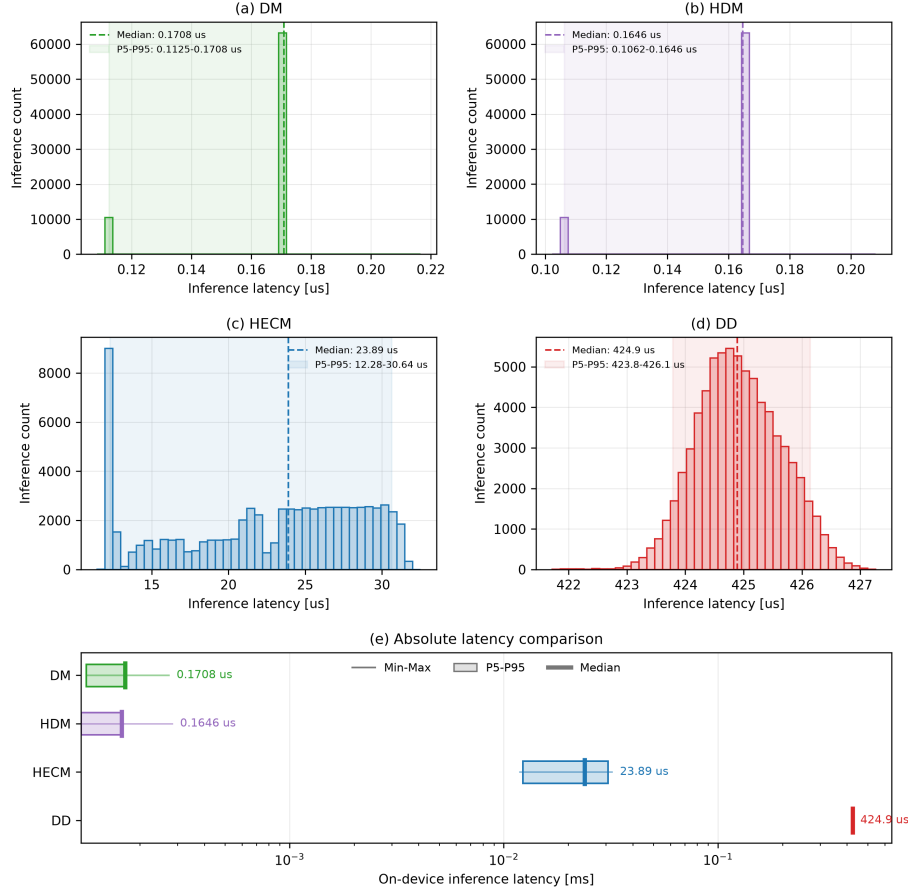


Figure 18: On-device SOC-core latency distributions across six cells and three replay rounds. Panels (a–d) show per-inference distributions. Panel (e) compares minimum, 5th percentile, median, 95th percentile, and maximum on a logarithmic scale. DD denotes continuous-state inference in this cross-model latency comparison.

682 it uniformly. HDM adapts the capacity denominator through learned SOH and  
 683 thereby reduces capacity mismatch, yet it retains the same charge-integration  
 684 core. Its similar response to current-gain error, missing charge information, and  
 685 ADC quantization follows directly from this shared structure.

686 HECM combines current-driven state propagation with a voltage residual

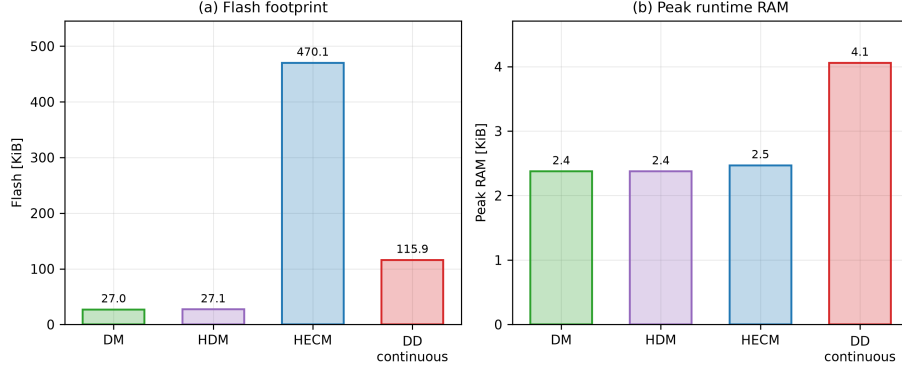


Figure 19: Embedded memory requirements of the isolated STM32 SOC implementations. (a) Compiled flash occupancy. (b) Peak runtime RAM measured during a representative holdout-test replay, combining statically allocated variable storage with maximum dynamic-memory allocation and call-stack use. DD denotes continuous-state execution, matching Figure 18. Exact rolling-window execution is compared separately in Figure 20.

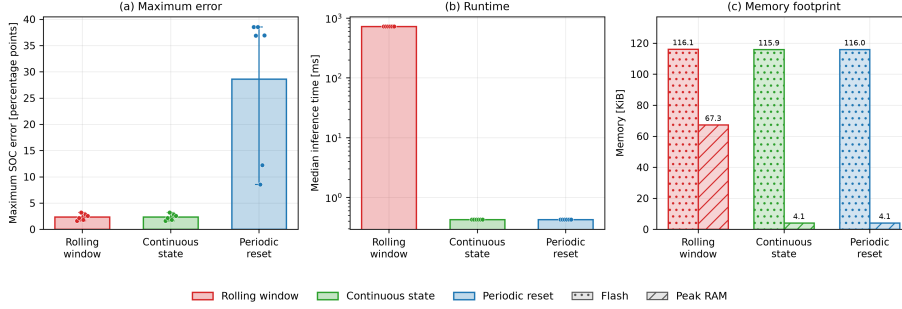


Figure 20: DD inference-mode trade-off on the STM32. The panels compare maximum dataset-SOC error, median latency, compiled flash occupancy, and measured peak runtime RAM for exact rolling-window, continuous-state, and periodically reset recurrent execution. Cell points show that periodic resets, not continuous forwarding, create the dominant accuracy loss.

687 and has the strongest observed recovery profile under the tested initialization  
 688 mapping. High current noise and burst dropout produce its largest point penal-  
 689 ties because corrupted or frozen inputs affect both state propagation and ob-  
 690 server correction. The confidence interval for high current noise includes zero  
 691 and shows substantial between-cell uncertainty. The dedicated lookup analysis

shows that the lower HECM current-gain penalty is retained under systematic  $\pm 10\%$  resistance scaling and  $\pm 10\text{ mV}$  OCV shifts. The largest absolute macro interaction is 0.00026 MAE. Because the same perturbations affect baseline accuracy more visibly, lookup calibration remains relevant even though the incremental current-gain response is locally stable. This table dependence and the associated flash footprint form the central deployment trade-off of HECM.

DD has the lowest nominal cell-macro error and the best adverse current-gain score in this comparison. It is also comparatively resilient to isolated sample loss. Event-aligned voltage spikes expose a recurrent transient that full-window MAE hides. The selected voltage and derivative features and the rolling recurrent context provide a plausible transmission path for this response and identify a concrete target for feature ablation and disturbance-aware training.

## 5.2. *Why nominal accuracy is insufficient*

Nominal MAE describes average agreement with the dataset ground truth under clean inputs. The disturbance and recovery experiments add the DD voltage-spike transient, HECM re-anchoring behavior, shared-SOH feature coupling, and model-specific responses to gain error and missing information. Embedded measurements add latency, memory, and inference-state semantics. These dimensions create a deployment profile for each implementation, while the weighting analysis shows how application priorities reshape their combined scores. Estimator selection can therefore begin with the relevant fault and recovery requirements and then incorporate the available compute, memory, and state-service architecture.

## 5.3. *Implications for BMS deployment*

Among the tested representatives, DD has the lowest nominal error against the dataset ground truth and remains a strong candidate when its recurrent implementation cost and voltage-spike response fit the application. HECM is preferable when persistent correction after a state mismatch has high priority and a larger flash image is available. DM and HDM remain attractive when

deterministic sub-microsecond execution and small memory dominate. Their integration structure places correspondingly greater requirements on current calibration, reliable sampling, and full-charge anchoring. These mechanism-based trade-offs connect the benchmark results directly to BMS deployment priorities.

The STM32 results connect algorithmic behavior with deployment cost. All four isolated SOC cores meet the one-second deadline in the tested replay, while their timing margins span more than six orders of magnitude between DM/HDM and exact rolling-window DD. Continuous-state DD removes most of the rolling-window cost by forwarding its recurrent state. The firmware measurements include flash, maximum call-stack use, dynamic-memory demand, and numerical equivalence. Integration of the shared SOH-LSTM, energy measurement, pack communication, and concurrent BMS scheduling forms the next system-level validation stage. Table 3 relates the measured estimator profiles to concrete deployment priorities.

Figure 16 complements the recommendation map by combining the measured accuracy, robustness, and recovery dimensions into relative priority profiles. The three profiles assign 60% to their named dimension and 20% to each remaining dimension. Family-balanced, equal-scenario, and high-severity aggregations produce different score distances and change the HDM-HECM order. This sensitivity turns the score into a configurable decision aid, while the scenario heatmap identifies the disturbance-specific strengths and weaknesses behind each profile.

#### 5.4. Limitations and future extensions

The conclusions are grounded in four specified implementations and the six-cell holdout test set from one controlled LFP aging dataset. Alternative reset logic, ECM structures, neural architectures, features, and training data can produce different estimator profiles. The low- and middle-load strata contain multiple cells, whereas the high-load stratum contains one exploratory trajectory. One low-load test trajectory also provides no mid-life or aged window.

Table 3: Decision-oriented recommendation map derived from the benchmark results. The table identifies the strongest observed representative under one dominant deployment priority and the stated protocol boundaries.

| Primary deployment priority and selected representative        | Main supporting benchmark evidence                                                                      | Main trade-off to consider                                                           |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Lowest nominal SOC error: DD                                   | Lowest six-cell macro MAE (0.0258) and RMSE (0.0300)                                                    | Rolling-window execution is the most computationally expensive implementation        |
| Strongest observed recovery from initialization mismatch: HECM | Lowest observed persistent-recovery-or-censor time and substantially fewer first-entry relapses than DD | Recovery is compared only after all estimators provide valid output                  |
| Lowest adverse current-gain penalty: DD                        | Matched signed sensitivity gives $\Delta$ MAE 0.0038 at 3%, below HECM, HDM, and DM                     | Temperature-offset and local voltage-spike sensitivity remain visible                |
| Smallest observed periodic/random sample-loss penalty: DD      | $\Delta$ MAE 0.0011/0.0026                                                                              | The duration-matched dropout effect is non-positive but highly variable across cells |
| Lowest deterministic compute cost: DM/HDM                      | Sub-microsecond median SOC-core latency and approximately 27 KiB flash                                  | Integration structure is sensitive to gain error and missing charge information      |
| Practical fast neural deployment: DD continuous state          | Approximately 0.425 ms median and 4.1 KiB peak runtime RAM                                              | Different recurrent semantics from the primary rolling-window benchmark              |

Figure A.21 makes these coverage limits explicit. The reported confidence intervals therefore quantify the available between-cell variation within this design. Transfer to other chemistries, cell formats, pack configurations, thermal environments, and field duty cycles requires corresponding cell-disjoint validation. Nominal accuracy is referenced to one consistently reconstructed offline SOC ground truth based on Coulomb counting and voltage anchors.

The primary cross-model campaign concentrates on the BMS measurement

chain, signal availability and timing, and deployment-accessible initialization states. The separate HECM lookup analysis probes uniform local resistance and OCV deviations, while broader parameter mismatch, pack-level interactions, and supervisory BMS faults form a separate validation layer. The initialization experiment maps an SOC-equivalent offset into each estimator’s available state and compares recovery once all estimators provide valid output. The resulting times characterize operational re-anchoring under class-specific interventions rather than identical latent-state convergence. The continuous high-load holdout replay complements the test-set windows with a mechanism-focused lifecycle view. The hourly SOH service represents gradual capacity fade, and channel-removal experiments would further isolate the DD voltage-spike transmission path. HECM uses HPPC-derived parameter surfaces identified from the model-development sets and held fixed throughout evaluation. The sensitivity analysis therefore characterizes local robustness around this calibrated model rather than arbitrary lookup-table error.

The hardware profile covers the isolated SOC firmware with SOH supplied as an external causal trace. Runtime RAM is measured during a representative holdout-test replay, including complete rolling-window DD inference. The next hardware stage combines the SOC and SOH services, measures energy demand, and evaluates estimator scheduling alongside communication and supervisory BMS tasks. Broader data campaigns can then extend the disturbance protocol to interacting faults, additional operating environments, and pack-level closed-loop behavior.

## 6. Conclusion

This study establishes a common robustness and embedded-performance benchmark for four structurally different SOC-estimation implementations. DD achieves the lowest nominal cell-macro error and the strongest results across several disturbance families, while local and hardware-specific analyses reveal its voltage-spike transient and rolling-window cost. HECM provides the strongest

787 observed recovery profile and voltage-residual correction, together with the  
788 largest flash demand and pronounced responses to current noise and prolonged  
789 dropout. Its lower current-gain penalty remains stable under the tested local  
790 resistance and OCV lookup perturbations, although lookup calibration con-  
791 tinues to affect nominal accuracy. DM and HDM provide deterministic sub-  
792 microsecond execution with minimal memory, while their current-integration  
793 structure increases sensitivity to gain error, missing charge information, and  
794 physical re-anchoring.

795 The combined evidence turns estimator selection into a deployment-oriented  
796 comparison of accuracy, disturbance-specific robustness, recovery, latency, flash,  
797 and runtime RAM. The STM32 measurements confirm numerical equivalence  
798 with the software implementations and one-second execution of all isolated SOC  
799 cores, while the DD deployment variants show how recurrent-state handling  
800 changes both cost and transient behavior. Scenario-level evidence remains cen-  
801 tral because the preferred implementation changes with the sensing fault profile,  
802 recovery requirement, and available hardware margin. This framework provides  
803 a direct basis for extending validation to complete SOC-SOH execution, concur-  
804 rent BMS tasks, additional operating environments, and pack-level applications.

## 805 **Appendix A. Dataset Split, Holdout Coverage, and Evaluation Pro-** 806 **TOCOL**

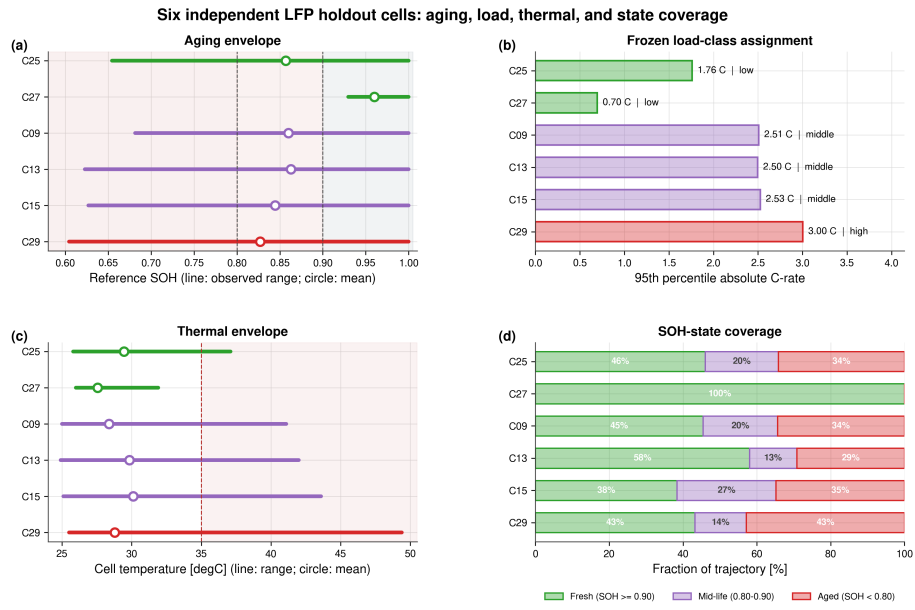


Figure A.21: Quantitative coverage of the six-cell independent holdout test set. Panels show observed SOH, the descriptive load-class assignment, thermal envelope, and full-trajectory SOH-state coverage. The high-load group contains one exploratory trajectory.

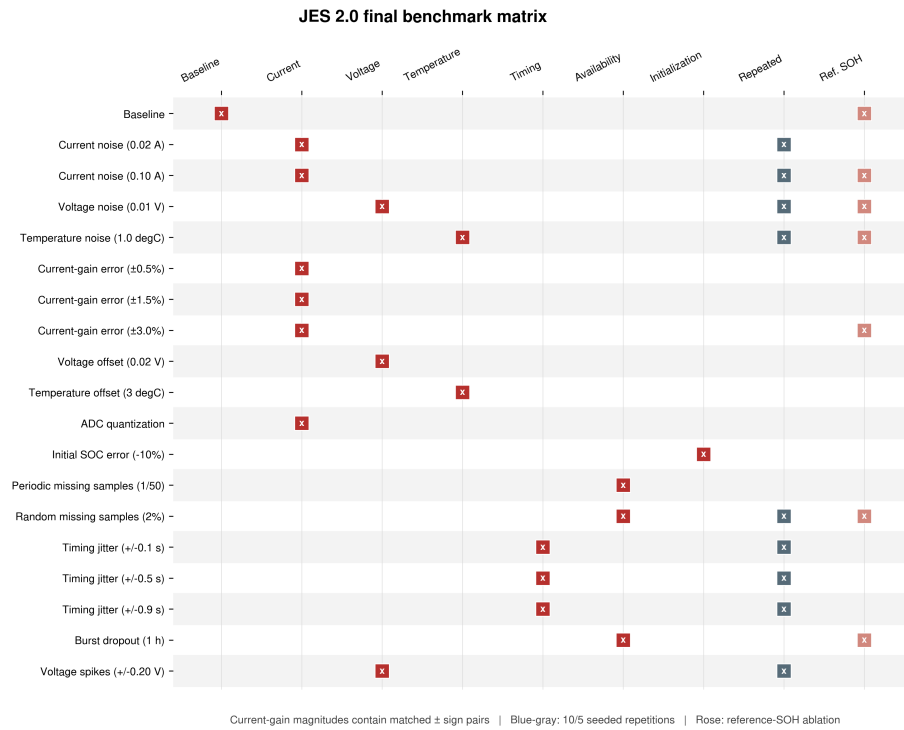


Figure A.22: Complete 19-scenario benchmark matrix. Red markers identify the manipulated measurement, timing, availability, or initialization channel. Blue-gray markers denote repeated stochastic cases. Rose markers identify paired reference-SOH ablations. Each current-gain magnitude contains matched positive and negative sign sublevels.

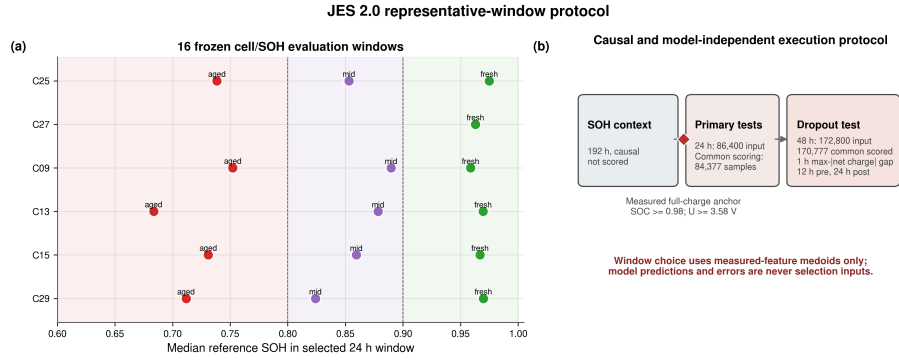


Figure A.23: Representative-window protocol. Sixteen cell/SOH windows are selected using measured-feature medoids. Every estimator is scored over the same interval once the rolling DD input context is complete. Standard windows cover 24 h after the causal SOH context, while the dropout case uses 48 h and places the one-hour gap at the eligible interval with maximum absolute net charge.

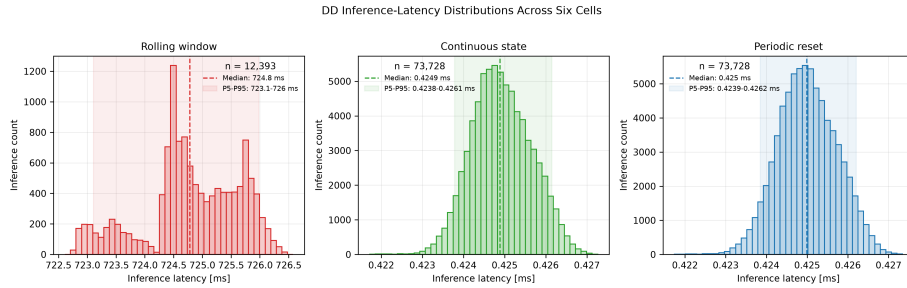


Figure A.24: DD latency distributions over the six hardware cell sequences. Exact rolling-window execution is shown separately from continuous-state and periodically reset execution because their scales and recurrent semantics differ substantially.

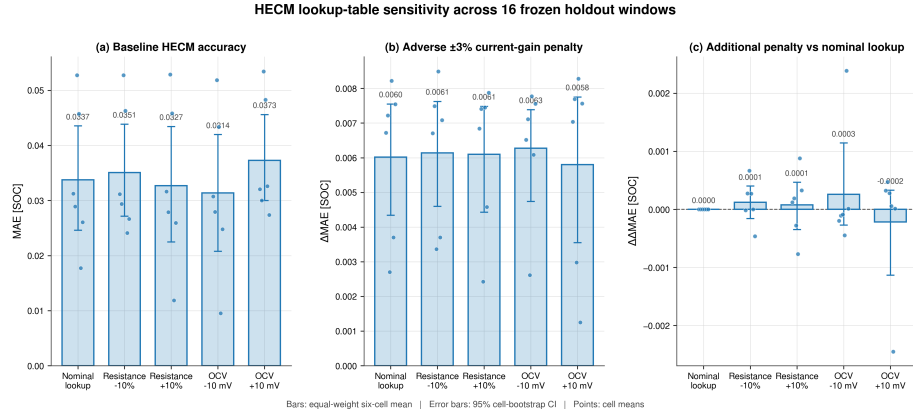


Figure A.25: HECM lookup-table sensitivity over all 16 frozen windows. (a) Baseline MAE under the nominal lookup, joint  $\pm 10\%$  scaling of  $R_i$ ,  $R_1$ , and  $R_2$ , and uniform  $\pm 10$  mV OCV shifts. (b) Adverse current-gain penalty from matched  $-3\%$  and  $+3\%$  gain errors relative to the baseline under the same lookup condition. (c) Lookup-gain interaction relative to the nominal-table penalty. Bars show equal-weight six-cell means, error bars show 95% cell-bootstrap intervals, and points show cell means.

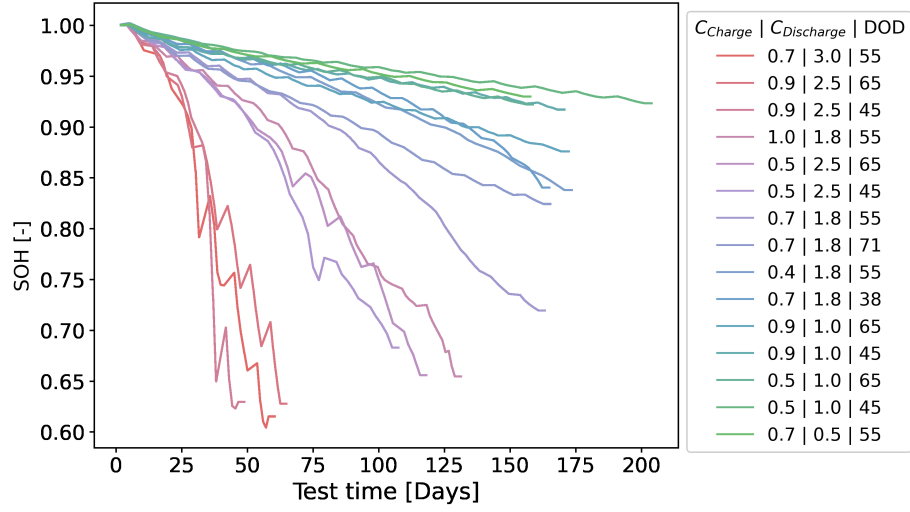


Figure A.26: Measured SOH trajectories across the complete LFP aging campaign. The campaign is divided into a six-cell training set, a three-cell validation set, and a six-cell independent holdout test set. The legend reports the applied charge C-rate, discharge C-rate, and depth of discharge for each trajectory. The plotted SOH values originate from repeated capacity checkups and are interpolated between checkup anchors. The exact cell-to-set assignment and measured coverage are reported in Table A.4.

Table A.4: Measured coverage and exact cell-to-set assignment for the fixed training, validation, and independent holdout test split. The 95th-percentile absolute C-rate is calculated over each complete 1-Hz trajectory. Load classes are defined only for the holdout test set. Its high-load class contains one cell and is exploratory.

| Cell | Split      | Duration<br>[d] | SOH range   | $T$ range<br>[°C] | P95<br>$ C_{\text{rate}} $ | Test-set load |
|------|------------|-----------------|-------------|-------------------|----------------------------|---------------|
| C01  | Training   | 198.6           | 0.922–1.000 | 24.3–29.4         | 1.01                       | –             |
| C03  | Training   | 153.3           | 0.921–1.000 | 25.9–30.7         | 1.00                       | –             |
| C05  | Training   | 166.1           | 0.917–1.000 | 25.2–30.6         | 1.00                       | –             |
| C11  | Training   | 113.2           | 0.656–1.000 | 25.6–42.8         | 2.48                       | –             |
| C17  | Training   | 158.7           | 0.719–1.000 | 25.8–34.9         | 1.75                       | –             |
| C23  | Training   | 167.8           | 0.837–1.000 | 26.1–37.2         | 1.74                       | –             |
| C07  | Validation | 167.6           | 0.874–1.000 | 25.2–32.0         | 1.01                       | –             |
| C19  | Validation | 160.4           | 0.840–1.000 | 26.3–33.9         | 1.75                       | –             |
| C21  | Validation | 160.8           | 0.824–1.000 | 26.2–37.6         | 1.75                       | –             |
| C09  | Test       | 102.6           | 0.682–1.000 | 25.0–41.1         | 2.51                       | Middle        |
| C13  | Test       | 44.7            | 0.623–1.000 | 24.9–42.0         | 2.50                       | Middle        |
| C15  | Test       | 60.7            | 0.627–1.000 | 25.1–43.6         | 2.53                       | Middle        |
| C25  | Test       | 127.1           | 0.655–1.000 | 25.8–37.1         | 1.76                       | Low           |
| C27  | Test       | 152.0           | 0.930–1.000 | 26.0–31.9         | 0.70                       | Low           |
| C29  | Test       | 56.0            | 0.604–1.000 | 25.5–49.4         | 3.00                       | High*         |

Table B.5: Six-cell baseline cell-macro statistics with hierarchical 95% confidence intervals.

| Estimator | MAE [SOC]               | RMSE [SOC]              | P95 error [SOC]         |
|-----------|-------------------------|-------------------------|-------------------------|
| DM        | 0.0690 [0.0499, 0.0823] | 0.0740 [0.0532, 0.0881] | 0.1088 [0.0768, 0.1307] |
| HDM       | 0.0412 [0.0301, 0.0515] | 0.0442 [0.0319, 0.0557] | 0.0649 [0.0449, 0.0846] |
| HECM      | 0.0337 [0.0246, 0.0434] | 0.0388 [0.0286, 0.0493] | 0.0644 [0.0468, 0.0821] |
| DD        | 0.0258 [0.0189, 0.0325] | 0.0300 [0.0223, 0.0366] | 0.0508 [0.0375, 0.0619] |

Table B.6: Exact paired baseline-MAE comparisons across the six-cell holdout test set. Differences are defined as estimator B minus estimator A. Holm correction is applied across the six model-pair tests.

| Estimator A | Estimator B | Mean difference<br>[SOC]   | 95% CI | $d_z$  | $p_{\text{exact}} / p_{\text{Holm}}$ |
|-------------|-------------|----------------------------|--------|--------|--------------------------------------|
| DM          | HDM         | -0.0278 [-0.0381, -0.0171] | -1.93  | 0.0313 | / 0.1875                             |
| DM          | HECM        | -0.0352 [-0.0470, -0.0224] | -2.00  | 0.0313 | / 0.1875                             |
| DM          | DD          | -0.0431 [-0.0558, -0.0286] | -2.29  | 0.0313 | / 0.1875                             |
| HDM         | HECM        | -0.0075 [-0.0162, 0.0015]  | -0.61  | 0.1563 | / 0.1875                             |
| HDM         | DD          | -0.0154 [-0.0243, -0.0084] | -1.39  | 0.0313 | / 0.1875                             |
| HECM        | DD          | -0.0079 [-0.0176, -0.0005] | -0.68  | 0.0938 | / 0.1875                             |

Table B.7: Selected cell-macro scenario penalties relative to paired baseline. Current-gain values use the adverse sign from the matched  $\pm 3\%$  sweep.

| Scenario                           | DM      | HDM     | HECM    | DD      |
|------------------------------------|---------|---------|---------|---------|
| Current noise, $\sigma_I = 0.10$ A | -0.0009 | +0.0000 | +0.0038 | -0.0000 |
| Current gain, adverse 3%           | +0.0108 | +0.0101 | +0.0060 | +0.0038 |
| Voltage offset, +0.02 V            | +0.0000 | +0.0005 | -0.0040 | +0.0038 |
| Temperature offset, +3 °C          | +0.0000 | +0.0002 | +0.0002 | +0.0074 |
| ADC quantization                   | +0.0038 | +0.0035 | -0.0018 | +0.0002 |
| Random missing samples, 2%         | +0.0080 | +0.0081 | +0.0039 | +0.0026 |
| Timing jitter, $\pm 0.9$ s         | -0.0003 | +0.0003 | +0.0006 | +0.0003 |
| Burst dropout, 1 h                 | +0.0050 | +0.0209 | +0.0239 | -0.0093 |
| Voltage spikes, $\pm 0.20$ V       | +0.0000 | -0.0003 | +0.0002 | -0.0001 |

Table B.8: Sensitivity of the illustrative normalized robustness score to the aggregation rule.

| Estimator | Family balanced | Equal scenario | Highest level plus offsets |
|-----------|-----------------|----------------|----------------------------|
| DM        | 0.562           | 0.574          | 0.615                      |
| HDM       | 0.468           | 0.504          | 0.512                      |
| HECM      | 0.428           | 0.560          | 0.576                      |
| DD        | 0.842           | 0.781          | 0.665                      |

Table B.9: Canonical paired initialization recovery and target-hardware summary. Recovery values are equal-weight cell means over the common evaluation interval. Endpoints already satisfied when all estimators first provide valid output are recorded as conservative upper bounds.

| Metric                                | DM    | HDM   | HECM  | DD                 |
|---------------------------------------|-------|-------|-------|--------------------|
| First 300-s entry/censor time [h]     | 22.28 | 17.21 | 1.12  | 1.29               |
| Persistent recovery/censor time [h]   | 22.28 | 22.28 | 1.20  | 2.60               |
| Recovery excess-error AUC [SOC h]     | 1.735 | 1.738 | 0.078 | 0.116              |
| First-entry censored fraction         | 0.89  | 0.67  | 0.00  | 0.00               |
| Persistent-recovery censored fraction | 0.89  | 0.89  | 0.00  | 0.00               |
| Relapse fraction after first hold     | 0.00  | 0.22  | 0.17  | 0.78               |
| Median SOC-core latency [ $\mu$ s]    | 0.171 | 0.165 | 23.9  | 424.9 <sup>a</sup> |
| Flash [KiB]                           | 27.0  | 27.1  | 470.1 | 115.9 <sup>a</sup> |
| Peak runtime RAM [KiB]                | 2.4   | 2.4   | 2.5   | 4.1 <sup>a</sup>   |

<sup>a</sup>Continuous-state DD. Exact rolling-window execution requires approximately 724 ms and 67.3 KiB peak RAM.

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## 812 **CRediT authorship contribution statement**

813 The author contribution statement is provided in the non-blinded submission  
814 materials.

## 815 **Declaration of competing interest**

816 The authors declare that they have no known competing financial interests or  
817 personal relationships that could have appeared to influence the work reported  
818 in this paper.

## 819 **Declaration of generative AI and AI-assisted technologies** 820 **in the manuscript preparation process**

821 During the preparation of this work the author(s) used ChatGPT, DeepL,  
822 and Grammarly in order to improve the linguistic quality and style of the  
823 manuscript. After using these tools, the author(s) reviewed and edited the  
824 content as needed and take(s) full responsibility for the content of the published  
825 article.

## 826 **Data availability**

827 The dataset is publicly available through its cited TU Berlin repository  
828 record [28]. Code, model configurations, benchmark definitions, random seeds,  
829 result summaries, hardware timing records, and analysis procedures are main-  
830 tained in a public GitHub repository. The repository contains the neural model  
831 assets and HECM parameterization used for the reported experiments.

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